Bellman-sufficient Information Complexity

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Abstract

We develop Bellman-sufficient information complexity, a representation-level framework for sequential decision making. The primitive benchmark is a fixed-truth environment space Ω with unrestricted nonanticipating algorithms. The intrinsic object is a Bellman-sufficient state representation, serving as an interactive notion of sufficient statistics, together with an information index $Y = \chi(\Omega)$, often the optimal decision or value object rather than the full environment. On the upper-bound side, learning is organized as a dynamic program on the sufficient state, equipped with a logarithmic information potential for the index. On the lower-bound side, a Bellman-Fano certificate uses the same state representation and information index, but propagates separate Bellman recursions for information gain and ghost mass. The central matching statement is therefore a conditional Bellman information-risk sandwich: when the log-penalized Bellman upper value and the ghost-quantile lower certificate close at the same radius, they certify the same complexity scale. Popular algorithms then appear as tractable certificates or relaxations of this common log-potential Bellman program, rather than as separate notions of information complexity.

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1 Introduction

Information-theoretic minimax theory asks how much risk remains when an experiment can reveal only limited information. In classical noninteractive estimation the experiment is fixed before the data are observed, and sharp rates are obtained by matching an upper information-risk construction with a lower entropy calculation: KL information measures what the experiment can distinguish, while local prior mass or packing entropy measures how many alternatives remain statistically indistinguishable. This is the perspective behind the information-risk upper and lower bounds of Zhang (2006), the entropy characterization of minimax rates in Yang and Barron (1999), and the classical Fano–Assouad method (Yu, 1997). Interactive decision making changes the experiment itself. The learner chooses actions by a nonanticipating policy, so the experiment itself is adapted to the past. Thus design, observation, and information accumulation are generated by one controlled process. The information calculation must therefore be dynamic: it should follow the Bellman recursion of the learning problem.

Algorithmic Fano's method is one way to respect this adaptivity. It compares the real history with an algorithm-dependent reference, or "ghost," history rather than with a fixed nonadaptive sampling scheme. This viewpoint appears in recent interactive Fano frameworks for general decision making (Chen et al., 2024) and in algorithmic lower bounds for representation learning (Xu, 2026); it is also closely related to the decision-estimation coefficient (DEC) approach of Foster et al. (2021) and its constrained or localized refinements (Foster et al., 2023). The central difficulty is that a sharp lower bound should not collapse a T-round adaptive trajectory into a one-round comparison too early. Such collapses require localized change-of-measure arguments because the action law under a hard environment need not match the action law under the reference experiment. They can hide the geometry of the evolving posterior, confidence set, or reference state, precisely where a matching upper bound must live.

The thesis of this paper is that the intrinsic object for sequential decision making is an indexed Bellman-sufficient representation. The frequentist environment space Ω, together with the unrestricted nonanticipating benchmark, identifies the primitive sequential decision-making objective. A Bellman-sufficient state representation turns the problem into a dynamic program over a sufficient state: a sufficient statistic, model posterior, frequentist estimator, or other representation that summarizes the environmental and historical information needed to close the Bellman recursion. The index $Y = \chi(\Omega)$ specifies the information target whose acquisition is charged. It may be the optimal action, an optimal policy, a value object, an active finite marginal, or the full environment.

The same formalism also explains why information potentials are not limited to statistical bandits. In planning, search, or reasoning, the latent instance may be a world model, problem instance, proof environment, or task specification, while the index may be a plan, answer, policy, value certificate, or proof object. Whenever the process admits a Bellman-sufficient state and a maintained marginal on that index, the logarithmic mass potential is again the generic information potential. This paper keeps the technical development in statistical learning and reinforcement learning, where calibration and Fano certificates can be stated precisely.

When a sufficient state representation closes the Bellman recursion with a smaller information index, the learner need not pay for estimating irrelevant features of the model. This is the sense in which learning is a special dynamic program: unlike ordinary computational dynamic programming, where the value function is problem-specific, the statistical potential is generically logarithmic in the index belief. The exact log-penalized Bellman program is the upper object that can match the Bellman-Fano ghost lower certificate; popular algorithms including UCB, E2D, and AMS/EBO are tractable certificates or relaxations of this same dynamic information principle.

The basic identity on the upper-bound side is formulated at a fixed frequentist truth and expressed through Bellman recursion and the chosen information index. Fix a truth $\omega^\star$, let $y^\star = \chi(\omega^\star)$, and let q_t be the retained reference marginal on the index. The specialized logarithmic potential

$$\phi_t(s; y^\star) = \gamma[-\log q_t(y^\star)]$$

turns the one-step fixed-truth log gain into a Bellman information potential. The corresponding indexed Algorithmic Information Ratio (AIR) bracket is

$$\ell_{\omega^\star}(S_t, p_t) + \mathbb{E}_{\omega^\star}[\phi_{t+1}(S_{t+1}; y^\star) \mid S_t, p_t] - \phi_t(S_t; y^\star),$$

and its sum telescopes to a regret bound of order $\gamma \log(1/q_1(y^\star))$ plus the accumulated bracket errors (Xu and Zeevi, 2025). Bayesian posterior averaging of the same coordinate identity gives the chain rule

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T} \sum_{t=1}^{T} \mathcal{I}_\chi(S'_t, p'_t),$$

where S'_t is the reference posterior state and p'_t is the algorithm's action distribution at that reference history. The key algorithmic principle is that the logarithmic mass penalty should be propagated through a controlled Bellman recursion.

The lower-bound side is organized by a Bellman-Fano certificate. The index specifies which uncertainty is charged, and the reference history specifies which low-regret ghost trajectories are counted. The algorithm-specific quantile theorem, Theorem 6.1, is an application of the interactive Fano method (Chen et al., 2024) to Bellman-sufficient representations; Theorem 6.3 is its algorithm-uniform Bellman-recursion certificate. In its simplest algorithm-specific form one obtains

$$\text{if} \qquad T\bar{C}_\chi^{\mathsf{Alg}}(\mu) \le \mathrm{kl}\left(\frac{1}{2}, p_r^{\mathsf{Alg}}(\mu, \chi)\right), \qquad \text{then} \qquad \mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) \ge \frac{Tr}{2}.$$

Here r is an average-regret threshold, p_r^{Alg} is the posterior-reference ghost probability of average regret at most r, and $\bar{C}_\chi^{\mathsf{Alg}} = T^{-1}\mathbb{E}\sum_t \mathcal{I}_\chi(S'_t, p'_t)$ is average indexed information. The critical balance is

$$T\bar{C}_\chi \asymp \log\frac{1}{p_r}.$$

When the ghost-good mass is roughly $\exp(-d_{\mathrm{eff}})$, the admissible total information is d_{eff}-level. This is the Fano/local-entropy radius, not the constant two-point radius. Theorem 6.3 converts the algorithmic lower-bound approach into a frequentist minimax statement. The certificate may use the exact Bellman-recursion values for information capacity and ghost mass, or valid supersolutions that upper-bound those exact values.

Theorem 3.4 states the resulting minimax information-risk sandwich. After a state, index, reference update, and calibration mechanism have been fixed, the upper side pays a log-penalized Bellman value with coordinate cost $\log(1/q_1(\chi(\omega)))$; the lower side pays a Bellman-Fano ghost entropy $\log(1/p_r)$. The central comparison is therefore the objective ratio between these two logarithmic quantities. When the ghost entropy has the same order as the upper code length, and the Bellman upper value has regular growth in its information budget, the upper and lower regret bounds match at the same radius. This is the interactive analogue of noninteractive information-risk matching: fixed-design local prior mass is replaced by ghost-good mass along an adaptive reference history, and fixed-sample KL is replaced by a controlled Bellman information telescope. We use "Bellman-sufficient information complexity" for this compatible state/index accounting. It becomes a matching complexity statement only after the upper and lower certificates are separately verified and shown to agree up to the stated constants or logarithmic factors.

The word "Bellman" in the title is deliberate. The exact value of an interactive problem is a dynamic program, following Bellman's principle of optimality (Bellman, 1957). An upper bound is a Bellman supersolution, or admissible relaxation in the sense of Rakhlin et al. (2012): it lies above the dynamic value and gives an algorithm. A lower bound is an exact Bellman-Fano value comparison, or a computable Bellman-Fano certificate obtained by upper-bounding the exact information-capacity and ghost-good-mass values. The canonical information-theoretic upper algorithm is therefore the AIR/MAIR information-potential Bellman program itself: it keeps the continuation value on the indexed state and optimizes the Bellman bracket before any one-step relaxation is imposed. UCB (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010), E2D (Foster et al., 2021, 2023), and AMS/EBO (Lattimore and György, 2021; Foster et al., 2022; Xu and Zeevi, 2025; Liu et al., 2025, 2026) are useful because they certify, relax, or robustify this Bellman bracket; they should not be confused with the exact Bellman program itself.

1.1 Main contributions.

1. We formulate *sequential decision making with Bellman-sufficient representations*, a representation-level framework for reinforcement learning and interactive decision making. In this framework, one studies the unrestricted nonanticipating benchmark through Bellman recursion, while a chosen index $Y = \chi(\Omega)$ specifies the decision-relevant information whose acquisition is charged. The state must close feasible actions, fixed-truth prediction, loss evaluation, and updating; when posterior-reference information is used, it must also close posterior predictive and conditional-index laws. Several bandit and reinforcement learning examples illustrate the framework with states given by sufficient statistics, model posteriors, frequentist estimators, or other Bellman-sufficient representations.

2. We prove the indexed posterior-reference chain rule and the generic indexed AIR regret identity. AIR is recovered by taking the decision index $Y = A^\star(\Omega)$, while MAIR is recovered by taking

the model or environment index $Y = \Omega$. We then record simplified statements and alternative proofs, show how the gradient bracket unifies different update rules, and explain the role of Danskin's theorem in the original AIR proof of Xu and Zeevi (2025).

3. We isolate the main information-complexity sandwich. The upper certificate pays an initial coordinate code length, while the Bellman-Fano lower certificate is governed by ghost entropy. We prove entropy-gap and regret-gap comparison statements under regularity conditions.

4. We give a unified "one identity, four algorithm families" account centered on a log-penalized Bellman upper theorem. The first family is the exact logarithmic information-potential Bellman program; this is the tightest upper object that can match the Bellman-Fano lower certificate. UCB methods certify the fixed-truth log-potential bracket through self-normalized calibration and optimism. E2D methods optimize a one-step offset obtained by replacing the continuation value by a local statistical separation penalty. AMS/EBO methods optimize KL-regularized robust belief relaxations, equivalently dual log-partition relaxations, of the same indexed AIR bracket.

5. We formulate the algorithmic quantile lower theorem and the Bellman-Fano lower certificates as algorithm-uniform minimax lower bounds. The lower-bound argument uses the same stepwise indexed information, state representation, and Bellman recursion to control both information gain and ghost mass. Bandit examples show that this approach recovers matching lower bounds.

6. We develop several representative examples, including multi-armed bandits, linear bandits, kernel bandits with an indexable action set, and conditional Bellman-rank embeddings. The kernel-bandit example highlights a regime in which the relevant active-action index is finite, even though the unknown RKHS function is infinite-dimensional. In such cases, optimism and its indexed AIR reinterpretation provide a natural mechanism for obtaining sharp complexity bounds, whereas global DEC optimization may be uninformative without appropriate localization or representation restriction.

1.2 Organization.

Section 2 defines sequential decision making, information indices, and Bellman-sufficient representations, and records the basic sufficient-statistic and full-posterior examples. Section 3 states the information-complexity sandwich and the entropy/regret comparison principles. Section 4 gives the indexed information telescope and the exact AIR/MAIR regret identities. Section 5 gives the log-penalized Bellman upper theorem and upper algorithmic relaxations including UCB families, E2D, and AMS/EBO. Section 6 gives the quantile lower theorem and the Bellman-Fano certificates. Section 7 provides applications to kernel bandits, multi-armed bandits, linear bandits, Bellman-rank embeddings, and estimator-based extensions to contextual bandits and reinforcement learning.

2 Sequential decision making with Bellman-sufficient representation

2.1 Sequential decision making and Bellman state compression

The primitive object is sequential decision making on histories. The framework encompasses both bandits and reinforcement learning in rich environments (Lattimore and Szepesvári, 2020; Sutton and Barto, 2018), as well as planning, search, reasoning, and optimal control (Silver and Sutton, 2025; Cormen et al., 2001; Bertsekas, 2017).

The sequential decision-making problem. There is an environment class Ω, and performance is required for every fixed truth $\omega^\star \in \Omega$. A prior μ is introduced only when we form a Bayes value, a Yao lower bound, a posterior-reference trajectory, or an algorithmic belief. We distinguish the raw action–observation trajectory from the full pre-decision information. The raw trajectory is

$$H_{t-1}^0 = (A_1, O_1, \ldots, A_{t-1}, O_{t-1}),$$

whereas $\bar{H}_{t-1}$ denotes the full information available just before the round-t action A_t. It may include the current context or physical state, public randomization, memory variables or estimators, or any other pre-decision variable that is genuinely available in the controlled experiment. The decision rule Alg itself is not a random component of the history: it is the object being optimized over. This notation makes the framework flexible enough to accommodate information that is not explicitly displayed in the raw trajectory, while also simplifying the notation used later. The central point, however, is simpler: from basic online learning models to dynamic planning models, future losses and observations generally depend on history.

At time t, after $\bar{h} \in \bar{\mathcal{H}}_{t-1}$, the learner may choose an action in a measurable set $\mathcal{A}_t(\bar{h})$. An unrestricted nonanticipating algorithm is a sequence of kernels

$$\pi_t^{\mathsf{Alg}}(\cdot \mid \bar{h}) \in \Delta(\mathcal{A}_t(\bar{h})), \qquad \bar{h} \in \bar{\mathcal{H}}_{t-1},$$

with deterministic algorithms included as degenerate kernels. Let $\mathfrak{A}_T^{\mathrm{na}}$ denote the class of all such algorithms. Unless an admissible subclass is explicitly named, the exact frequentist objective in this paper is

$$\mathfrak{R}_T^\star(\Omega) := \inf_{\mathsf{Alg} \in \mathfrak{A}_T^{\mathrm{na}}} \sup_{\omega \in \Omega} \mathbb{E}_\omega^{\mathsf{Alg}} \sum_{t=1}^{T} \ell_t(\omega, \bar{H}_{t-1}, A_t).$$

This is the reinforcement-learning and interactive-decision-making problem in its general sequential risk-minimization form: choose a nonanticipating algorithm whose Bellman risk is small uniformly over environments.

Action, observation, and decision clock. At time t, the learner samples $A_t \sim \pi_t^{\mathsf{Alg}}(\cdot \mid \bar{H}_{t-1})$. Under truth $\omega^\star$, the observation is drawn from

$$O_t \sim P_{\omega^\star, t}(\cdot \mid \bar{H}_{t-1}, A_t),$$

and the one-step cost, loss, or regret increment is $\ell_t(\omega^\star, \bar{H}_{t-1}, A_t)$. If the environment or adversary observes the announced mixed action $p_t = \pi_t^{\mathsf{Alg}}(\cdot \mid \bar{H}_{t-1})$ before the realized action, then p_t is treated as part of the current control and the primitives are written as $P_{\omega,t}(\cdot \mid \bar{H}_{t-1}, p_t, A_t)$ and $\ell_t(\omega, \bar{H}_{t-1}, p_t, A_t)$. The simpler notation suppresses this argument when the law and loss depend only on the realized action. The increment may be signed, provided the cumulative target used in a lower-bound theorem is integrable and has the stated lower bound, usually $L_\omega(H_T) \geq 0$. The decision clock t is an abstract Bellman clock. In a contextual bandit, the current context is part of $\bar{H}_{t-1}$ before the action is chosen. In episodic reinforcement learning, one may either take a macro-round view, where A_t is a whole policy and O_t is the resulting episode trajectory, or a micro-round view, where the decision clock contains both the outer episode index and the inner horizon stage.

Bellman state and Bellman sufficiency. Our theory centers on the specification of an analytical Bellman state for sequential decision making. This object should not be confused with the physical state often used in Markov decision process models. A compressed Bellman state is a dynamic sufficient statistic for decision making,

$$S_t = \phi_t(\bar{H}_{t-1}),$$

or equivalently $S_t = \phi_t(H^0_{t-1}, R_t)$ when the retained object R_t records the pre-decision information not displayed in the raw trajectory. Such a compression may be used in Bellman recursions only if the available actions, observation law, loss, and update are all measurable with respect to the retained state. Thus there must exist an action set $\mathcal{A}_t(s)$, an observation kernel $P_{\omega,t,s,a}$, and a loss function $\ell_{\omega,t}(s,a)$ such that, whenever $S_t(\bar{h}) = s$,

$$\begin{aligned} \mathcal{A}_t(\bar{h}) &= \mathcal{A}_t(s), \\ P_{\omega,t}(\cdot \mid \bar{h}, a) &= P_{\omega,t,s,a}, \\ \ell_t(\omega, \bar{h}, a) &= \ell_{\omega,t}(s,a). \end{aligned}$$

In addition, there must be an update map or controlled transition τ_t such that S_{t+1} is generated from the retained state and the current interaction, written in the basic notation as $S_{t+1} = \tau_t(S_t, A_t, O_t)$. If the one-step law or cost also depends on the announced mixed action, the same definition is used with (p, a) in place of a, for example $P_{\omega,t,s,p,a}$, $\ell_{\omega,t}(s,p,a)$, and $S_{t+1} = \tau_t(S_t, p, A_t, O_t)$. If any of these conditions fails, the retained state is incomplete and must be augmented; see Definition 2.2 for the formal statement. The full pre-decision history $S_t = \bar{H}_{t-1}$ always satisfies the condition. As illustrated in Section 2.5, a full environment posterior together with the current time/context/physical state also provides a fallback sufficient state for broad classes of Bayesian reference experiments.

The notation $P_{\omega,t,s,a}$ and $\ell_{\omega,t}(s,a)$ means only that all observed variables needed to determine the next observation law and the loss have been included in the Bellman state s. In applications this state may include an estimator or posterior object, the relevant physical state, an adversary state, or a relaxation variable. It does not mean that a bandit arm or an MDP transition kernel literally depends on an estimator alone. Once a Bellman-sufficient state is fixed, all main Bellman operators optimize over $p \in \Delta(\mathcal{A}_t(s))$. In stationary examples, or when the stage t is included in s, we sometimes write $\mathcal{A}(s)$ for $\mathcal{A}_t(s)$.

Reference dynamic program and minimax lower bound. On a sufficient Bellman state, with update $S_{t+1} = \tau_t(S_t, A_t, O_t)$, the Bayes/reference dynamic program induced by a posterior b_s is

$$V_{T+1}(s) = 0, \qquad V_t(s) = \inf_{p \in \Delta(\mathcal{A}_t(s))} \left\{ \ell(s,p) + \mathbb{E}_{a\sim p,\, o \sim P_{s,a}} V_{t+1}(\tau_t(s,a,o)) \right\}, \tag{1}$$

where

$$\ell(s,p) = \mathbb{E}_{a\sim p} \int \ell_\omega(s,a) b_s(d\omega), \qquad P_{s,a} = \int P_{\omega,s,a} b_s(d\omega).$$

Equation (1) is not the definition of the frequentist problem; it is the Bellman recursion induced by a chosen reference representation. Minimax lower bounds are connected to such reference recursions through Yao's principle,

$$\mathfrak{R}^\star_T(\Omega) \ge \sup_{\mu \in \Delta(\Omega)} \inf_{\mathsf{Alg} \in \mathfrak{A}^{\mathrm{na}}_T} \mathbb{E}_{\Omega\sim\mu} \mathbb{E}_{\mathbb{P}^{\mathsf{Alg}}_\Omega} \sum_{t=1}^T \ell_t(\Omega, \bar{H}_{t-1}, A_t).$$

The same log-potential Bellman calculus also applies to planning, search, and reasoning whenever those tasks can be cast as controlled experiments with a fixed latent instance, a retained state, and an index such as a plan, proof, answer, or value certificate. The paper focuses on statistical learning and reinforcement learning because these settings make both sides of the information-risk sandwich explicit: upper bounds are logarithmic-potential Bellman supersolutions, while lower bounds are Bellman-Fano ghost certificates.

Remark 2.1 (Adversarial and estimated losses)**.** *The framework is not restricted to stochastic regret losses. A nonstochastic or adaptive adversary may be treated as part of the fixed environment ω, provided its rule is nonanticipating. If the adversary observes only the past pre-decision history, the one-step primitives may be written as $\ell_t(\omega, \bar{H}_{t-1}, a)$ and $P_{\omega,t}(\cdot \mid \bar{H}_{t-1}, a)$. Classical online learning with adaptive adversaries sometimes also permits the adversary to observe the learner's announced mixed action p_t (Cesa-Bianchi and Lugosi, 2006). In that case, p_t is part of the Bellman state, and the corresponding primitives take the form $\ell_t(\omega, \bar{H}_{t-1}, p_t, a)$ and $P_{\omega,t}(\cdot \mid \bar{H}_{t-1}, p_t, a)$. This special setting remains compatible with the Bellman-state representation framework, although it lies slightly beyond the focused sequential decision-making model studied in the main development. Estimated-loss upper bounds can be handled by replacing the true loss in the Bellman bracket with a state-measurable surrogate or domination certificate, while carrying the resulting approximation error, as discussed in Sections 7.4 and 7.6. Lower bounds require the target cumulative indexed loss used in the ghost event to be lower bounded, usually nonnegative after a harmless shift.*

2.2 Information index, conditional loss, and entropy accounting

The latent environment space is Ω, but the information charged by the upper and lower bounds may be a coarser coordinate. An information index is a measurable map

$$Y = \chi(\Omega) \in \mathcal{Y}.$$

The index is chosen by the analyst and should reflect the decision-relevant object: an optimal arm, an optimal policy, a value certificate, a finite active marginal, or, when no compression is justified, the full environment. The fixed-truth Bellman recursion still uses $\omega^\star$. The index enters through the logarithmic potential, the posterior/reference information accounting, and the lower ghost event.

The loss need not be a function of Y alone. For Bayes or Fano calculations we therefore use the conditional index loss, assuming the relevant regular conditional laws exist,

$$\ell_t^\chi(y, \bar{h}, a) := \mathbb{E}_\mu\left[\ell_t(\Omega, \bar{h}, a) \mid \chi(\Omega) = y, \bar{H}_{t-1} = \bar{h}\right]. \tag{2}$$

If the current control includes an announced distribution p, then the same definition is used with $\ell_t^\chi(y, \bar{h}, p, a)$. When the retained state is an exact posterior-indexed lift in Definition 2.2, this conditional loss and the corresponding conditional predictive law are state-measurable and may be written as $\ell_{t,y}^\chi(s, a)$ and $P_{t,s,a}^y$. These conditional-index objects are not required for ordinary fixed-truth Bellman control; they are required for indexed posterior averaging and for the Bellman-Fano lower certificate.

We use the cumulative and average losses

$$L_\omega(H_T) = \sum_{t=1}^T \ell_t(\omega, \bar{H}_{t-1}, A_t), \qquad \bar{L}_\omega(H_T) = T^{-1} L_\omega(H_T),$$

and

$$L_\chi(y, H_T) = \sum_{t=1}^T \ell_t^\chi(y, \bar{H}_{t-1}, A_t), \qquad \bar{L}_\chi(y, H_T) = T^{-1} L_\chi(y, H_T).$$

Under the Bayesian mixture generated by $\Omega \sim \mu$,

$$\mathbb{E}_{\Omega\sim\mu, H_T\sim\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) = \mathbb{E}_{Y,H_T} L_\chi(Y, H_T). \tag{3}$$

For the upper identities, the increments may be any integrable costs for which the displayed expectations exist. For the quantile lower bounds, the object that must be nonnegative is the cumulative indexed loss, not necessarily each one-step increment. Throughout the clean lower-bound statements we assume $L_\chi(Y, H_T) \geq 0$ almost surely under the true mixture. If a problem has a known lower bound $L_\chi \geq -B$, the same statements apply to the shifted loss $L_\chi + B$, with the corresponding shift subtracted at the end.

The two logarithmic quantities compared later are the upper coordinate code length and the lower ghost entropy. If q_1 is the initial reference marginal on $\mathcal{Y}$, the fixed-truth upper telescope pays

$$L_0(\Omega_0; q_1, \chi) := \sup_{\omega\in\Omega_0} \log \frac{1}{q_1(\chi(\omega))}.$$

If q_1 is uniform on a finite index set of size M, then $L_0 = \log M$. At a regret radius r, the Bellman-Fano lower theorem uses the ghost-good probability

$$p_r = \mathbb{P}_{Y\sim q_1, H'_T}\{\bar{L}_\chi(Y, H'_T) \leq r\}, \qquad E_r := \log \frac{1}{p_r}.$$

Thus the primitive entropy comparison is L_0/E_r, not the size of the full model class. The detailed comparison and its conversion into a regret gap are stated in Section 3.

2.3 Bellman Sufficiency versus classical sufficiency and Bellman rank

We now formally state the representation condition used by the rest of the paper. The frequentist truth $\omega^\star$ remains fixed. Priors, posteriors, confidence objects, exponential-weights distributions, and algorithmic beliefs are reference objects: they may define algorithms or certificates, but they are not substitutes for the fixed environment unless a theorem explicitly takes a Bayesian average.

Definition 2.2 (Indexed Bellman-sufficient representation)**.** *Fix an index map $\chi : \Omega \to \mathcal{Y}$. A retained state process*

$$S_t = \phi_t(\bar{H}_{t-1})$$

(equivalently $S_t = \phi_t(H^0_{t-1}, R_t)$ when R_t records retained pre-decision information) is a fixed-truth Bellman-sufficient representation *if the following objects are determined by the current time-state pair (t, s).*

(i) Admissible actions. *There is a state-measurable action set $\mathcal{A}_t(s)$ such that $\mathcal{A}_t(\bar{h}) = \mathcal{A}_t(s)$ whenever $S_t(\bar{h}) = s$.*

(ii) Predictive sufficiency. *For every environment ω and action $a \in \mathcal{A}_t(s)$, there is a kernel $P_{\omega,t,s,a}$ such that, for every pre-decision history $\bar{h}$ with $S_t(\bar{h}) = s$,*

$$P_{\omega,t}(\cdot \mid \bar{h}, a) = P_{\omega,t,s,a}.$$

(iii) Loss sufficiency. *There is an integrable state-measurable cost function $\ell_{\omega,t}(s, a) \in \mathbb{R}$ such that*

$$\ell_t(\omega, \bar{h}, a) = \ell_{\omega,t}(s, a) \qquad \textit{whenever } S_t(\bar{h}) = s.$$

The lower-bound quantile theorem does not require every increment to be nonnegative; it requires the cumulative indexed loss to be lower bounded, and in the displayed lower bounds we assume $L_\chi(Y, H_T) \geq 0$ almost surely.

(iv) Update sufficiency. *There is an update map or controlled kernel τ_t such that the next retained state is generated from the retained state and the current interaction. In the deterministic-update notation used in most displays,*

$$S_{t+1} = \tau_t(S_t, A_t, O_t),$$

with the evident replacement $\tau_t(S_t, p, A_t, O_t)$ when p is part of the one-step control. Random next contexts or physical next states are included in O_t; randomization internal to a reference update may be included in the pre-decision information or represented by an equivalent transition kernel.

When these clauses hold, Bellman recursion is closed under each fixed truth. A compressed state is therefore a dynamic sufficient statistic for the Bellman primitives: feasible actions, prediction, loss evaluation, and continuation.

If, in addition, a prior or reference law μ is fixed and one wants state-based posterior-reference information accounting, the state must also carry the relevant index and posterior-predictive objects. The coarsened state posterior is

$$b^S_{t,s} := \mathcal{L}_\mu(\Omega \mid S_t = s), \qquad q^S_{t,s} := \chi_\# b^S_{t,s},$$

and, for $q^S_{t,s}$-almost every y,

$$b^{S,y}_{t,s} := \mathcal{L}_\mu(\Omega \mid S_t = s, \chi(\Omega) = y).$$

An exact posterior-indexed lift on the state *requires the posterior predictive and index-conditional objects*

$$P^S_{t,s,a} := \int P_{\omega,t,s,a}\, b^S_{t,s}(d\omega), \qquad P^{S,y}_{t,s,a} := \int P_{\omega,t,s,a}\, b^{S,y}_{t,s}(d\omega), \tag{4}$$

and

$$\ell^{\chi,S}_{t,y}(s,a) := \int \ell_{\omega,t}(s,a) b^{S,y}_{t,s}(d\omega) \tag{5}$$

to be functions of (t, s, a, y), together with a state-measurable reference update for $q^S_{t,s}$. These posterior-predictive and conditional-index clauses are not needed for the fixed-truth Bellman recursion itself. They are needed when the posterior-averaged chain rule, the Bayes information-potential program, or the Bellman-Fano lower certificate is run on the compressed state.

The notation will usually suppress the explicit time subscript when the stage is part of the state. Thus $P_{\omega,s,a}$, $\ell_\omega(s,a)$, and $\mathcal{A}(s)$ mean $P_{\omega,t,s,a}$, $\ell_{\omega,t}(s,a)$, and $\mathcal{A}_t(s)$ at the current Bellman time. This convention is harmless only after the state includes the variables that make actions, losses, observation laws, and updates state-measurable.

Relationship and distinction to classical sufficient statistics and Bellman rank. Classical sufficiency provides the static prototype for Bellman-state sufficiency (Fisher, 1922): a statistic $T(X)$ is sufficient for a family $\{P_\theta\}$ when the conditional distribution of the data given T is independent of θ. Bellman sufficiency is a closely related but conceptually distinct controlled analogue. The retained state must preserve everything needed for future actions, predictions, losses, information coordinates, and updates under admissible controls. In particular, one need not estimate all of $\omega^\star$ if a smaller state closes these recursions.

As explained above, however, Bellman sufficiency is a closedness condition for the Bellman recursion, not a requirement of classical exact sufficiency expressed through conditional independence. The exact state-based lift in Definition 2.2, namely the fifth axiom and equations (4)–(5), should therefore be distinguished from classical model sufficiency, index-marginal sufficiency, or posterior sufficiency with respect to the full pre-decision history. The Bellman-sufficiency clauses do not impose such conditional-independence statements, nor are they automatically implied by them. If these clauses fail, the resulting state-based posterior/reference process is only a coarsened information accounting. It remains a valid certificate only when the corresponding upper Bellman inequality or lower Bellman–Fano recursion is proved for that coarsened state. In particular, one cannot obtain a lower information complexity merely by discarding information without also closing the relevant Bellman inequalities. This distinction is important for the existential debate in Section 2.6, which explains why sufficient states matter and why they can be tighter, or more intrinsic, than using the full pre-decision history without compression.

The index $Y = \chi(\Omega)$ is the coordinate whose logarithmic mass is charged by the upper information potential and whose ghost-good mass appears in the lower Fano certificate. It need not identify the whole environment. In bandits it may be the optimal arm; in reinforcement learning it may be an optimal policy or value object; in planning or reasoning it may be a plan, proof, answer, or certificate. The mathematics is the same whenever a Bellman-sufficient state and a reference marginal q_s for the index are available.

The formalism is aligned with the representation-level philosophy behind low Bellman rank (Jiang et al., 2017), but it is deliberately stricter. Bellman rank is a factorization of expected Bellman errors relative to a function class, roll-in distribution, and witness family. It is not, by itself, a state: it does not determine the realized next observation, the fixed-truth loss, the posterior or reference update, or the conditional laws for an index. It becomes an indexed Bellman-sufficient representation only after those missing objects are supplied and shown to close the recursions above.

The following basic examples belong in the framework section because they verify Definition 2.2 directly. They also fix the interpretation of the state used later: the fixed truth remains the environment, while sufficient statistics, posteriors, and finite marginals are retained Bellman states or reference objects.

2.4 Basic examples of Bellman-sufficient representation

Structured bandit problems offer particularly transparent examples of the Bellman-sufficiency framework for sequential decision making, which is guided by the classical intuition of sufficient statistics.

Example 2.3 (Classical sufficient statistics as one-step Bellman states)**.** Consider a nonadaptive dominated experiment with observations $X_1, \dots, X_n$ from P_θ, and no control. Let $T_n = T(X_1, \dots, X_n)$ satisfy the Fisher–Neyman factorization. If the loss and index depend on θ only through quantities whose posterior distribution is determined by T_n, then the process $S_t = (t, T_{t-1})$ is Bellman sufficient. Predictive sufficiency is the usual conditional factorization; update sufficiency is $T_t = u(T_{t-1}, X_t)$; posterior predictive and index sufficiency follow because the posterior or conditional reference law depends on the history only through T_t. In a regular exponential family,

$$p_\theta(x) = h(x) \exp\{\vartheta(\theta)^\top T(x) - A(\theta)\},$$

with conjugate or otherwise statistic-measurable reference law, the cumulative statistic $\sum_{i<t} T(X_i)$ gives the corresponding state. This is the static prototype for all examples below.

Example 2.4 (Finite stochastic multi-armed bandits)**.** Let $\Omega \subseteq \Theta^K$ and let arm a produce an observation from P_{θ_a}. For unit-variance Gaussian arms, $P_{\theta_a} = N(\theta_a, 1)$. A reference Gaussian prior with independent coordinates gives the posterior state

$$S_t = \big(t, (n_{t,a}, \bar{X}_{t,a}, v_{t,a})_{a=1}^K\big),$$

where $n_{t,a}$ is the number of previous pulls of arm a, $\bar{X}_{t,a}$ is the empirical mean when $n_{t,a} > 0$, and $v_{t,a}$ is the posterior variance. Equivalently one may store the posterior hyperparameters. For a fixed truth $\omega = \theta$,

$$P_{\omega,s,a} = N(\theta_a, 1), \qquad \ell_\omega(s,a) = \max_b \theta_b - \theta_a,$$

and the update of $(n, \bar{X}, v)$ after observing arm a is a function of (s, a, o). The posterior predictive law is $N(m_{t,a}, 1 + v_{t,a})$, and for the action index $Y = A^\star(\theta)$ the conditional predictive law $P^y_{s,a}$ and conditional loss $\ell^\chi_y(s,a)$ are obtained by integrating the same Gaussian posterior over the event $A^\star(\theta) = y$. Hence Definition 2.2 is satisfied. The empirical mean alone is not sufficient: the count or posterior variance is needed for prediction, confidence, and information gain.

Example 2.5 (Finite-dimensional Gaussian linear bandits)**.** Let $\Omega \subseteq \mathbb{R}^d$, actions satisfy $\|a\|_2 \le 1$, and

$$O_t = \langle \theta, A_t \rangle + \xi_t, \qquad \xi_t \sim N(0,1).$$

Under a Gaussian reference prior, the exact Bayesian state is $S_t = (t, m_t, \Sigma_t)$, where

$$\Sigma_{t+1}^{-1} = \Sigma_t^{-1} + A_t A_t^\top, \qquad m_{t+1} = \Sigma_{t+1}\{\Sigma_t^{-1} m_t + A_t O_t\}.$$

For fixed θ, $P_{\theta,s,a} = N(\langle \theta, a\rangle, 1)$ and the regret loss is $\sup_{\|b\|_2 \le 1} \langle \theta, b\rangle - \langle \theta, a \rangle$. The posterior predictive law is $N(\langle m_t, a\rangle, 1 + a^\top \Sigma_t a)$, and conditioning the Gaussian posterior on an index event such as $\chi(\theta) = \theta$, $\chi(\theta) = \theta/\|\theta\|_2$, or $\chi(\theta) = A^\star(\theta)$ determines $P^y_{s,a}$ and $\ell^\chi_y(s,a)$. Therefore the Gaussian mean–covariance pair is an exact Bellman-sufficient state. A frequentist UCB state $(\hat{\theta}_t, V_t, \beta_t)$ is different: it can certify a fixed-truth bracket on the calibration event $\|\hat{\theta}_t - \theta^\star\|_{V_t} \le \beta_t$, but it is not an exact posterior-reference state unless a reference belief or confidence-to-belief map is added.

Example 2.6 (Finite active marginals in kernel bandits)**.** Let $f^\star$ belong to an RKHS on a possibly infinite domain, but suppose the learner is evaluated on a finite active set $\mathcal{X} = \{x_1, \ldots, x_n\}$. Under the GP reference law on the active reward vector $F = (f(x_1), \ldots, f(x_n))$, the finite marginal posterior $(m_t, \Sigma_t) \in \mathbb{R}^n \times \mathbb{R}^{n\times n}$ is Bellman sufficient for the finite experiment. For fixed $f^\star$, pulling x_i has law $N(f^\star(x_i), \lambda)$ in the Gaussian reference calculation; the posterior predictive law is $N(m_{t,i}, \lambda + \Sigma_{t,ii})$; and the usual rank-one Gaussian update is a function of (m_t, Σ_t, i, O_t). If the index is the optimal active action $Y = \arg\max_i F_i$, then index sufficiency follows by conditioning the finite Gaussian posterior on $\arg\max_i F_i = y$. The infinite-dimensional RKHS function is still the frequentist truth; the sufficient state is finite only because the decisions and observations use the active marginal.

2.5 Full environment posterior is fallback sufficient state

For sequential decision making, a natural Bellman-sufficient representation is obtained by retaining the full environment posterior. For suitably defined Markovian model classes over the state space, which include standard episodic model-based reinforcement learning settings, this posterior serves as a fallback sufficient state.

Example 2.7 (Full environment posterior as fallback sufficient state)**.** *Consider a realizable model class in which each environment $\omega \in \Omega$ specifies, for every retained physical state or context z, action a, and stage t, an observation kernel $P_{\omega,t}(\cdot \mid z, a)$, a loss $\ell_{\omega,t}(z, a)$, and the physical-state transition rule. Let Z_t denote the physical state, context, stage, or episode coordinate needed to make these kernels Markov. Fix a reference prior μ and set*

$$\Pi_t = \mathcal{L}_\mu(\Omega \mid \bar{H}_{t-1}), \qquad S_t = (Z_t, \Pi_t).$$

Assume the observation laws are dominated on the relevant support. Then S_t gives an exact Bellman-sufficient state. Indeed, for each fixed truth ω,

$$P_{\omega,S_t,a} = P_{\omega,t}(\cdot \mid Z_t, a), \qquad \ell_\omega(S_t, a) = \ell_{\omega,t}(Z_t, a),$$

and the next retained state is obtained by updating the physical coordinate and the posterior

$$\Pi_{t+1}(d\omega) = \frac{p_{\omega,t}(O_t \mid Z_t, A_t)\Pi_t(d\omega)}{\int p_{\omega',t}(O_t \mid Z_t, A_t)\Pi_t(d\omega')}.$$

For an index $\chi : \Omega \to \mathcal{Y}$, the indexed marginal and conditional predictive components are

$$q_t = \chi_{\#}\Pi_t, \qquad \Pi_t^y = \mathcal{L}_{\Pi_t}(\Omega \mid \chi(\Omega) = y),$$

$$P^y_{S_t,a} = \int P_{\omega,t}(\cdot \mid Z_t, a)\, \Pi_t^y(d\omega), \qquad \ell_y^\chi(S_t, a) = \int \ell_{\omega,t}(Z_t, a)\, \Pi_t^y(d\omega).$$

Thus prediction, loss evaluation, posterior/reference updating, and indexed information accounting all close on S_t. This verifies Definition 2.2.

In this sense, sequential decision making with Bellman-sufficient representation contains the Decision Making with Structured Observations (DMSO) framework (Foster et al., 2021) as a special case when DMSO is interpreted as an independent episodic model-learning formalism and the full environment posterior is retained as the state. At the same time, the present framework also accommodates Markovian dependence across physical states and inner horizon stages, as in computational dynamic programming. The more important point is not formal inclusion, but the possibility of running dynamic-programming certificates on sufficient states and of charging a decision-relevant index rather than the full model label.

Example 2.8 (DMSO as a posterior-state specialization)**.** *In the independent-episode DMSO formalism Foster et al. (2021), an environment M specifies the observation law and loss for each decision rule within an episode, and episodes are conditionally independent given M. The raw episode history and the full pre-decision history carry the same statistical information about M: for a fixed prior and likelihood, the posterior $\Pi_k = \mathcal{L}(M \mid \bar{H}_{k-1})$ is a deterministic compression of the observations available before episode k. Taking Ω to be the DMSO model class and retaining Π_k, together with any public episode context or admissible-decision constraints, gives a literal Bellman-sufficient state for the reference experiment. The posterior predictive observation law, the posterior update after the episode, and every index marginal $\chi_{\#}\Pi_k$ are functions of this state. Thus DMSO fits the present framework as the special case in which the Bellman clock is the outer episode and the posterior is the retained state coordinate. More compressed representations require an additional sufficiency proof; they do not follow from the DMSO notation alone.*

If the adversarial model sequence depends only on the pre-decision history $\bar{H}_{k-1}$, and not on the learner's current randomization, then adversarial DMSO (Foster et al., 2022; Xu and Zeevi, 2025) is directly subsumed by the Bellman-state representation framework. The same conclusion also holds for adversaries that observe the current decision rule through the extension described in Remark 2.1. In either case, one may take the full environment to be the entire nonanticipating model sequence, with the posterior over this environment serving as a valid posterior state.

2.6 Why sufficient states and compression matter

The full pre-decision history $\bar{H}_{t-1}$ is always a valid Bellman state, so sufficiency is not needed for formal existence of a dynamic program. Its purpose is existential and representational. A useful state is a quotient of the full history on which the Bellman recursion, the logarithmic information potential, and the Bellman-Fano ghost calculation all close. Such a quotient need not be unique, nor must it be minimal.

It is already clear that sufficient states are useful at a practical level: specifying them makes it tractable to identify the relevant information index, construct upper-bound algorithms, and formulate lower-bound certificates. The question in this subsection is more conceptual: whether sufficient states offer any information-theoretic advantage beyond the full pre-decision history. The answer is yes. By data processing, compression can only reduce the information budget; when the retained state is Bellman-sufficient in the appropriate controlled sense, this reduction need not lose any of the Bellman structure required for upper or lower bounds. The key is to distinguish Bellman sufficiency from classical conditional-independence sufficiency. Otherwise, if the relevant information gain were always invariant under compression, sufficient states would be more a methodological convenience than an intrinsic representation-level object. A general formal claim that every proof device operating on the full pre-decision history, but not passing through the constructed sufficient state, must lead to coarser bounds would be too strong. We do not pursue such a claim here.

It is helpful to separate three notions. Classical model sufficiency is the static conditional-independence condition

$$\Omega \perp \bar{H}_{t-1} \mid S_t,$$

which says that the state retains all posterior information about the full environment. Index-marginal sufficiency for $Y = \chi(\Omega)$ is weaker:

$$Y \perp \bar{H}_{t-1} \mid S_t.$$

It says only that the state retains the full-history posterior marginal of the index. Bellman sufficiency, as used in Definition 2.2, is different: it is the dynamic closure of the primitives needed for decision making, namely feasible actions, fixed-truth predictive kernels, losses or surrogate losses, updates, and, when needed, the indexed reference objects used by the logarithmic potential and the Bellman-Fano certificate. Thus a Bellman-sufficient state need not be sufficient for the full model, and a state-based reference process may be coarser than the full-history posterior; its validity comes from the Bellman inequalities that close on that state.

Proposition 2.9 (Bellman quotient and information monotonicity). *Let $S_t = \phi_t(\bar{H}_{t-1})$ be a fixed-truth Bellman-sufficient representation. Suppose a one-step Bellman cost c_ω and a continuation value F_{t+1} are state-measurable, so that*

$$c_\omega(\bar{h}, p) = c_\omega(s, p), \qquad F_{t+1}(\bar{H}_t) = f_{t+1}(S_{t+1}), \qquad s = S_t(\bar{h}).$$

Then the full-pre-decision-history Bellman operator factors through S_t: for every $\bar{h}$ with $S_t(\bar{h}) = s$,

$$\inf_{p\in\Delta(\mathcal{A}_t(\bar{h}))} \sup_{\omega\in\mathcal{C}_t(\bar{h})} \left\{c_\omega(\bar{h}, p) + \mathbb{E}_{\omega,\bar{h},p} F_{t+1}(\bar{H}_t)\right\} = \inf_{p\in\Delta(\mathcal{A}_t(s))} \sup_{\omega\in\mathcal{C}_t(s)} \left\{c_\omega(s, p) + \mathbb{E}_{\omega,s,p} f_{t+1}(S_{t+1})\right\},$$

whenever the comparison set is also state-measurable. Moreover, for any reference law and any fixed index $Y = \chi(\Omega)$,

$$I(Y; S_t) \le I(Y; \bar{H}_{t-1}).$$

Equality in the information display holds precisely under index-marginal sufficiency, $Y \perp \bar{H}_{t-1} \mid S_t$.

Proof. The Bellman-operator identity follows directly from the state-measurability of the action set, comparison set, one-step cost, transition law, and continuation value. Histories in the same fiber of S_t induce the same local optimization problem. The information inequality is the data-processing inequality applied to the measurable map $S_t = \phi_t(\bar{H}_{t-1})$. The equality condition is the standard equality case for conditional mutual information, $I(Y; \bar{H}_{t-1} \mid S_t) = 0$. □

This proposition is the formal reason sufficient states are useful. The quotient preserves the Bellman problem whenever the primitives close on the state, while data processing ensures that the state-based information accounting for a fixed index is no larger than the full-pre-decision-history accounting. If exact index-marginal sufficiency holds, the two information quantities are equal; if the state intentionally uses a coarser algorithmic or reference belief, the information quantity may be strictly smaller. That strict reduction is not a free theorem about the original unrestricted problem: it is valid only when the upper Bellman certificate and the lower Bellman-Fano recursion are both proved on the same coarsened state. In this sense the framework is existential and representational. One searches for a state and index on which the dynamic program and the entropy accounting simultaneously close; when they do, the resulting sandwich can be tighter and more intrinsic than a certificate written directly on the full pre-decision history, even for the same global index Y.

3 Information complexity sandwich

This section isolates the comparison that the rest of the paper uses. Once a Bellman-sufficient state, index, reference update, and calibration mechanism have been fixed, the upper and lower bounds are expressed in the same units. The upper logarithmic Bellman theorem pays an initial coordinate code length for the realized index (Theorem 5.2). The lower Bellman-Fano theorem is governed by the ghost entropy of reference histories that are already good for the sampled index (Theorem 6.3). The exact lower-side quantities are themselves Bellman values: an indexed-information capacity value and a ghost-good-mass value. Computable proofs may replace these exact values by valid supersolutions, but the intrinsic comparison is between the upper code length and the exact or certified ghost entropy obtained from the lower Bellman recursions. The only generic statistical mismatch is therefore the ratio between these two logarithmic quantities; all remaining gaps come from relaxation, calibration, optimization, or from growth of the Bellman upper value as the information budget changes.

3.1 Code length, ghost entropy, and localization

Let $\Omega_0 \subseteq \Omega$ be the problem class being lower and upper bounded, let $\chi : \Omega \to \mathcal{Y}$ be the index, and let q_1 be the initial reference marginal on $\mathcal{Y}$. The fixed-truth upper telescope pays the worst-case initial coordinate code length (Section 5)

$$L_0(\Omega_0; q_1, \chi) := \sup_{\omega \in \Omega_0} \log \frac{1}{q_1(\chi(\omega))}, \tag{6}$$

with the convention that the value is infinite if $q_1(\chi(\omega)) = 0$ for some $\omega \in \Omega_0$. For a lower-bound prior whose index marginal is q_1, and for an average-regret radius r, define the ghost-good mass and ghost entropy by

$$p_r := \mathbb{P}_{Y \sim q_1,\, H'_T}\{\bar{L}_\chi(Y, H'_T) \le r\}, \qquad E_r := \log \frac{1}{p_r}. \tag{7}$$

Here H'_T is the reference history used in the Bellman-Fano certificate and is independent of Y under the ghost law. In the notation of Section 6, $p_r = p_r^{\mathsf{Alg}}(\mu, \chi)$ for an algorithm-specific theorem. For an algorithm-uniform theorem, the intrinsic ghost mass is the exact Bellman value $\Gamma_1^{r,\star}(q_1, s_1, 0)$, while a proof may use any valid upper certificate $\overline{\Gamma}_1^r(q_1, s_1, 0) \geq \Gamma_1^{r,\star}(q_1, s_1, 0)$. The certified entropy is then $-\log \overline{\Gamma}_1^r$, which is conservative but always valid.

Proposition 3.1 (Entropy gap between the log-Bellman upper and Bellman-Fano lower)**.** *Assume $E_r > 0$. The generic entropy mismatch between the fixed-truth logarithmic Bellman upper certificate and the Bellman-Fano lower certificate at radius r is*

$$\kappa_r := \frac{L_0(\Omega_0; q_1, \chi)}{E_r}.$$

If $\mathcal{Y}_0 = \chi(\Omega_0)$ is finite with $|\mathcal{Y}_0| = M$ and q_1 is uniform on $\mathcal{Y}_0$, then $L_0 = \log M$. If the ghost-good mass is merely bounded away from one, say $p_r \leq p_0 < 1$, then $E_r \geq \log(1/p_0)$ and the crude finite-index entropy ratio is at most $\log M / \log(1/p_0)$. Thus a logarithmic gap follows from a nonvacuous lower certificate, but constant-factor matching requires a ghost entropy of order $\log M$, not merely a constant ghost entropy. Moreover, if for every reference history h' the set

$$G_r(h') := \{y \in \mathcal{Y}_0 : \bar{L}_\chi(y, h') \leq r\}$$

has cardinality at most m_r, then

$$E_r \geq \log \frac{M}{m_r}, \qquad \kappa_r \leq \frac{\log M}{\log(M/m_r)}.$$

In particular, constant-factor entropy matching holds whenever the ghost history is r-good for at most $M^{1-o(1)}$ indices in a way that makes $\log(M/m_r) \asymp \log M$.

Proof. The fixed-truth upper theorem pays $\log(1/q_1(\chi(\omega)))$ for truth ω, hence at most L_0 uniformly over Ω_0. The Bellman-Fano lower theorem is controlled by the logarithmic inverse ghost-good probability $E_r = \log(1/p_r)$. This proves the definition of κ_r. If $p_r \leq p_0 < 1$, then $E_r \geq \log(1/p_0)$, giving the crude logarithmic finite-index comparison. Under the uniform finite-index law,

$$p_r = \mathbb{E}_{H'_T} \frac{|G_r(H'_T)|}{M} \leq \frac{m_r}{M}.$$

Taking logarithms gives the stated lower bound on E_r and the corresponding upper bound on κ_r. $\square$

3.2 From entropy comparison to regret comparison

An entropy comparison alone is not yet a regret comparison. The missing ingredient is a growth property of the upper Bellman value as a function of its information budget, analogous to the condition used in the constrained DEC complexity-sandwich theory of Foster et al. (2023). For a fixed state/index representation and the log-penalized Bellman program in Section 5.1, write

$$\mathsf{U}_T(L) := \inf_{\gamma > 0} \left\{ W_1^\gamma(s_1) + \gamma L + \Delta_T^\gamma \right\}, \tag{8}$$

where W^γ is the log-penalized Bellman value and $\Delta_T^\gamma \geq 0$ collects worst-case calibration, approximation, and optimization errors for that certificate as in Theorem 5.2. For the exact calibrated Bellman dynamic program, $\Delta_T^\gamma = 0$. The function U_T is nondecreasing in L.

Assumption 3.2 (Bellman upper-growth condition)**.** *There exist constants $C_{\mathrm{gr}} \ge 1$ and $\beta \in [0,1]$ such that, for all $L > 0$ and $a \ge 1$,*

$$\mathsf{U}_T(aL) \le C_{\mathrm{gr}} a^{\beta} \mathsf{U}_T(L).$$

The assumption is a regularity condition on the upper certificate, not a theorem about every sequential problem. It is the Bellman analogue of the growth or sub-root conditions used when converting entropy radii into risk radii in fixed experiments.

Assumption 3.3 (Bellman-Fano lower certification at radius r)**.** *For the same state/index representation, suppose a prior μ supported on Ω_0, with $\chi_{\#}\mu = q_1$, induces exact or certified lower Bellman quantities C_{low} and E^r_{low}. Here C_{low} is an upper bound on the algorithm-uniform indexed information accumulated along posterior-reference histories, and E^r_{low} is a lower bound on the ghost entropy, equivalently $p_r \le e^{-E^r_{\mathrm{low}}}$. The radius r is certified if*

$$C_{\mathrm{low}} + \log 2 \le \frac{1}{2} E^r_{\mathrm{low}}.$$

The intrinsic choice is $C_{\mathrm{low}} = C^{\star}_1(s_1)$ and $E^r_{\mathrm{low}} = -\log \Gamma^{r,\star}_1(q_1, s_1, 0)$, where $C^{\star}$ and $\Gamma^{r,\star}$ are the exact lower Bellman values of Section 6.3. Supersolutions give computable conservative choices $C_{\mathrm{low}} = \overline{C}_1(s_1)$ and $E^r_{\mathrm{low}} = -\log \overline{\Gamma}^r_1(q_1, s_1, 0)$.

Theorem 3.4 (Frequentist Bellman information-risk sandwich)**.** *Fix $\Omega_0 \subseteq \Omega$, a horizon T, an index χ, a Bellman-sufficient state/reference update, and an initial index law q_1 with finite $L_0 = L_0(\Omega_0; q_1, \chi)$. Suppose the upper side admits a calibrated log-penalized Bellman certificate whose budget value is U_T in (8). Then*

$$\mathfrak{R}^{\star}_T(\Omega_0) \le \mathsf{U}_T(L_0). \tag{9}$$

Conversely, suppose Assumption 3.3 holds at radius r with lower information certificate C_{low} and ghost entropy certificate $E^r_{\mathrm{low}} > 0$. Then

$$\mathfrak{R}^{\star}_T(\Omega_0) \ge \frac{Tr}{2}. \tag{10}$$

Consequently, if in addition Assumption 3.2 holds and the same radius satisfies the base closing condition

$$\mathsf{U}_T(E^r_{\mathrm{low}}) \le C_{\mathrm{base}} Tr, \tag{11}$$

then the upper and lower bounds satisfy

$$\mathfrak{R}^{\star}_T(\Omega_0) \le C_{\mathrm{gr}} C_{\mathrm{base}} \left(\max\left\{1, \frac{L_0}{E^r_{\mathrm{low}}}\right\}\right)^{\beta} Tr, \tag{12}$$

and hence the regret-ratio gap relative to the lower bound $Tr/2$ is at most

$$2C_{\mathrm{gr}} C_{\mathrm{base}} \left(\max\left\{1, \frac{L_0}{E^r_{\mathrm{low}}}\right\}\right)^{\beta}.$$

For a finite uniform index, Proposition 3.1 gives the corresponding entropy ratio whenever the certified ghost entropy is obtained from the exact ghost mass, or from a valid ghost-mass upper bound satisfying the same multiplicity estimate: $L_0/E^r_{\mathrm{low}} \le \log M / \log(M/m_r)$ under localization, and the cruder $L_0/E^r_{\mathrm{low}} = \log M/\log(1/p_r)$ for the exact ghost probability without localization. If only $p_r \le p_0 < 1$ is known, the ratio is at most $\log M/\log(1/p_0)$; this is logarithmic index-cardinality, not a constant-factor match.

Proof. The upper inequality is Theorem 5.2, optimized over γ, with the uniform code length bounded by L_0 and the remaining certificate errors included in Δ_T^γ. The lower inequality is Theorem 6.3 together with Yao's principle, since μ is supported on Ω_0. For the final comparison, set $a = \max\{1, L_0/E^r_{\text{low}}\}$. Since U_T is nondecreasing and $aE^r_{\text{low}} \geq L_0$,

$$\mathsf{U}_T(L_0) \leq \mathsf{U}_T(aE^r_{\text{low}}) \leq C_{\text{gr}} a^\beta \mathsf{U}_T(E^r_{\text{low}}) \leq C_{\text{gr}} C_{\text{base}} a^\beta Tr.$$

Combining this with $\mathfrak{R}^\star_T(\Omega_0) \geq Tr/2$ gives the displayed regret-ratio bound. □

Remark 3.5 (What the comparison does and does not say)**.** *The finite-index gap is not automatically* $\log|\chi(\Omega)|$*, and it is not automatically constant. The objective entropy gap is* L_0/E^r_{low}*. If the ghost-good mass is only bounded away from one, so that the lower ghost entropy is merely constant, then the crude upper–lower comparison may indeed be logarithmic in the index size. If the ghost mass is localized so that* $E^r_{\text{low}} \asymp L_0$*, the entropy gap is constant. This is why multi-armed bandits and the hypercube linear-bandit lower bound can match the log-Bellman upper scale: the ghost-good histories are good for only a small fraction of the possible indices. In poorly localized problems, a one-step DEC coefficient or a global model-index entropy may leave a logarithmic model-cardinality or worse gap; the gap reflects the certificate, not an intrinsic impossibility of the Bellman-sufficient representation.*

4 Indexed information and exact AIR/MAIR identities

4.1 Posterior-reference histories and index information

The mutual-information identities in this subsection use the exact posterior-reference version of Definition 2.2. The fixed-truth log-gain regret identity in the next subsection should not be substituted into the posterior chain rule unless the two objects are induced by a coherent reference law.

For a prior μ and algorithm Alg, define the Bayesian posterior-reference trajectory law

$$\bar{\mathbb{P}}^{\mathsf{Alg}}_\mu := \int \mathbb{P}^{\mathsf{Alg}}_\omega \, \mu(d\omega).$$

A reference history is denoted

$$H'_T \sim \bar{\mathbb{P}}^{\mathsf{Alg}}_\mu, \qquad S'_t = \phi_t(H'_{t-1}, b'_t), \qquad b'_t = \mathcal{L}_\mu(\Omega \mid H'_{t-1}),$$

and the algorithm's reference decision kernel is

$$p'_t = p_t(\cdot \mid H'_{t-1}).$$

Definition 4.1 (Indexed information gain)**.** *At a current time-state* (t,s)*, let* q_s *be the posterior law of* $Y = \chi(\Omega)$ *and let* $P^y_{s,a}$ *and* $P_{s,a}$ *be the conditional and marginal predictive laws from Definition 2.2; we suppress* t *in these laws when the stage is part of* s*. For* $p \in \Delta(\mathcal{A}_t(s))$*, define*

$$\mathcal{I}_\chi(s,p) := \mathbb{E}_{a\sim p} \int D_{\text{KL}}(P^y_{s,a} \| P_{s,a}) q_s(dy). \tag{13}$$

When $Y = \Omega$*, this is model-index information. When* $Y = \pi^\star(\Omega)$*, this is action-index information. In the action-index case, the belief is still a belief over environments; only the information target is the optimal action.*

Define the cumulative and average indexed information of an algorithm by

$$C_\chi^{\mathsf{Alg}}(\mu) := \mathbb{E}_{H_T'\sim\bar{\mathbb{P}}_\mu^{\mathsf{Alg}}} \sum_{t=1}^{T} \mathcal{I}_\chi(S_t', p_t'), \qquad \bar{C}_\chi^{\mathsf{Alg}}(\mu) := \frac{1}{T} C_\chi^{\mathsf{Alg}}(\mu). \tag{14}$$

Proposition 4.2 (Reference-history indexed chain rule)**.** *Assume that the retained state is an exact posterior-indexed lift in the sense of Definition 2.2, and that the displayed KL terms are well defined and integrable. For every prior μ, index map χ, and algorithm* Alg*,*

$$I_\mu(Y; H_T) = \mathbb{E}_{H_T'\sim\bar{\mathbb{P}}_\mu^{\mathsf{Alg}}} \sum_{t=1}^{T} \mathcal{I}_\chi(S_t', p_t') = T\bar{C}_\chi^{\mathsf{Alg}}(\mu). \tag{15}$$

Proof. The mutual-information chain rule gives

$$I_\mu(Y; H_T) = \sum_{t=1}^{T} I_\mu(Y; A_t, O_t \mid H_{t-1}).$$

Given H_{t-1}, the action A_t is sampled by the algorithm using only randomization independent of the environment and is conditionally independent of Y. Therefore

$$I_\mu(Y; A_t, O_t \mid H_{t-1}) = I_\mu(Y; O_t \mid H_{t-1}, A_t).$$

Conditioning further on $A_t = a$, the law of O_t given $Y = y$ is $P_{S_t,a}^y$, while the posterior predictive law is $P_{S_t,a}$. Hence

$$I_\mu(Y; O_t \mid H_{t-1}, A_t = a) = \int D_{\mathrm{KL}}(P_{S_t,a}^y \| P_{S_t,a}) q_{S_t}(dy).$$

Averaging over $a \sim p_t(\cdot \mid H_{t-1})$ and over the marginal law of the history under the Bayesian mixture law proves the result. Writing this marginal law as the reference history law gives the displayed expression. □

4.2 Regret identity for sequential decision making

The information gain in (13) is posterior-averaged. The regret identity used in frequentist upper bounds is slightly finer: it fixes the true environment and tracks the posterior log ratio of its index.

Fix a state s, an environment ω, and let $y = \chi(\omega)$. Assume domination and $q_s(y) > 0$. Define

$$\Lambda_\chi(s, a, o; y) := \log \frac{dP_{s,a}^y}{dP_{s,a}}(o).$$

For $p \in \Delta(\mathcal{A}_t(s))$, define the fixed-truth index log gain

$$J_\chi(s, p; \omega) := \mathbb{E}_{a\sim p,\, O\sim P_{\omega,s,a}} \Lambda_\chi(s, a, O; \chi(\omega)). \tag{16}$$

Equivalently,

$$J_\chi(s, p; \omega) = \mathbb{E}_{a\sim p} \left[D_{\mathrm{KL}}(P_{\omega,s,a} \| P_{s,a}) - D_{\mathrm{KL}}(P_{\omega,s,a} \| P_{s,a}^{\chi(\omega)}) \right]. \tag{17}$$

Thus the fixed-truth log gain is a pointwise-in-ω quantity, whereas $\mathcal{I}_\chi(s, p)$ is a posterior-averaged index information. More explicitly, for each index value y with positive posterior mass,

$$\mathbb{E}_{\omega\sim\mu_s(\cdot|\chi(\omega)=y)} J_\chi(s, p; \omega) = \mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{s,a}^y \| P_{s,a}),$$

and averaging this display over $y \sim q_s$ gives $\mathcal{I}_\chi(s,p)$. Besides this conditional posterior averaging, a special model-index $\chi(\omega) = \omega$ case gives $P_{\omega,s,a} = P_{s,a}^{\chi(\omega)}$ and

$$J^{\mathrm{MAIR}}(s,p;\omega) = \mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{\omega,s,a} \| P_{s,a}),$$

In this sense, $J_\chi(s,p;\omega)$ serves as a common fixed-truth log-gain object for stating both algorithmic information gain and fixed-truth estimation error.

Coefficient convention. Throughout the upper-bound identities and Bellman programs, the information multiplier is denoted by $\gamma > 0$: a one-step log gain G is charged as γG, and a coordinate code length is charged as $\gamma \log(1/q)$. Some AIR/MAIR/DEC papers use a temperature parameter η with multiplier $1/\eta$ (Xu and Zeevi, 2025; Liu et al., 2025, 2026); in the present notation that convention is simply $\gamma = 1/\eta$.

For $\gamma > 0$, define the fixed-truth indexed bracket

$$B_{\chi,\gamma}(s,p;\omega) := \ell_\omega(s,p) - \gamma J_\chi(s,p;\omega), \qquad \ell_\omega(s,p) := \mathbb{E}_{a\sim p} \ell_\omega(s,a). \tag{18}$$

Theorem 4.3 (Regret identity for sequential decision making)**.** *Suppose the algorithm chooses an H_{t-1}-measurable decision distribution $p_t \in \Delta(\mathcal{A}_t(S_t))$ and then samples $A_t \sim p_t$. Fix $\omega^\star$ and $y^\star = \chi(\omega^\star)$. Assume that $q_t(y^\star) > 0$ and that the relevant likelihood ratios are well defined along the analyzed trajectory. If the reference index posterior is updated by Bayes' rule after the realized action and observation, then, conditionally on H_{t-1},*

$$\ell_{\omega^\star}(S_t,p_t) = B_{\chi,\gamma}(S_t,p_t;\omega^\star) + \gamma \mathbb{E}_{A_t\sim p_t,\, O_t \sim P_{\omega^\star,S_t,A_t}} \left[\log \frac{q_{t+1}(y^\star)}{q_t(y^\star)} \middle| H_{t-1} \right]. \tag{19}$$

Consequently,

$$\mathbb{E}_{\omega^\star}^{\mathsf{Alg}} L_{\omega^\star}(H_T) = \gamma \mathbb{E}_{\omega^\star}^{\mathsf{Alg}} \log \frac{q_{T+1}(y^\star)}{q_1(y^\star)} + \mathbb{E}_{\omega^\star}^{\mathsf{Alg}} \sum_{t=1}^{T} B_{\chi,\gamma}(S_t,p_t;\omega^\star). \tag{20}$$

The expectation is over the action randomization, the observation process, and any exogenous randomness of the algorithm. Thus the identity is an equality in expectation; a pathwise equality is available only after replacing the randomized loss by its realized counterpart and keeping the corresponding martingale difference.

Proof. For the realized action $A_t = a$, Bayes' rule for the reference index posterior gives

$$\log \frac{q_{t+1}(y^\star)}{q_t(y^\star)} = \log \frac{dP_{S_t,a}^{y^\star}}{dP_{S_t,a}}(O_t).$$

Taking conditional expectation under the fixed truth $\omega^\star$, first over $O_t \sim P_{\omega^\star,S_t,a}$ and then over $a \sim p_t$, yields

$$\mathbb{E}_{A_t\sim p_t,\, O_t \sim P_{\omega^\star,S_t,A_t}} \left[\log \frac{q_{t+1}(y^\star)}{q_t(y^\star)} \middle| H_{t-1} \right] = J_\chi(S_t,p_t;\omega^\star).$$

Rearranging the definition of $B_{\chi,\gamma}$ proves the one-step identity. Summing over t and taking expectation telescopes the posterior log ratio. The equality between expected realized loss and $\mathbb{E} \sum_t \ell_{\omega^\star}(S_t,p_t)$ follows from conditional expectation over $A_t \sim p_t$. □

The following corollary gives the deterministic specialization of Theorem 4.3, which will be useful in the subsequent UCB analyses. It follows directly from conditional determinism and the conditional-expectation step in (19). In this case the outer expectation over randomized decision laws drops out, leaving only the posterior-ratio telescope and the fixed-truth gradient bracket.

Corollary 4.4 (Deterministic-action indexed regret identity)**.** *Suppose the algorithm is deterministic after conditioning on H_{t-1}, so $p_t = \delta_{a_t}$. Fix $\omega^\star$ and $y^\star = \chi(\omega^\star)$. If the reference index posterior is updated by Bayes' rule, then*

$$\mathbb{E}^{\mathsf{Alg}}_{\omega^\star} L_{\omega^\star}(H_T) = \gamma \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \log \frac{q_{T+1}(y^\star)}{q_1(y^\star)} + \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \sum_{t=1}^{T} B_{\chi,\gamma}(S_t, \delta_{a_t}; \omega^\star).$$

The outer expectation is only over the observation process and any exogenous randomness. In a deterministic noiseless experiment the same identity is pathwise.

4.3 Exact AIR and MAIR regret identities

We now present simplified statements and alternative proofs of several results that are central to the AIR/MAIR literature (Xu and Zeevi, 2025; Liu et al., 2025, 2026). These results are best understood as regret identities, rather than merely as upper bounds. The results in this subsection can be understood as specializations and consequences of Theorem 4.3. In principle, the AIR/MAIR identities require only the Bellman-sufficient representation of Definition 2.2. We nevertheless formulate them in the DMSO language, both to align with earlier work and to simplify notation. Under this specialization, the index may be an optimal-action index or a model index. The state can be taken to be the full model posterior as illustrated in Section 2.5.

Let $\mathcal{M}$ be the model class and $\mathcal{A}$ be the decision space. Model M has reward mean $r_M(a)$ and observation law $P_{M,a}$. For a candidate optimal action $a^\star$, write

$$\Delta_{M,a^\star}(p) := r_M(a^\star) - \mathbb{E}_{a\sim p} r_M(a).$$

AIR. An AIR belief is a pair belief $\nu \in \Delta(\mathcal{M} \times \mathcal{A})$ over $(M, A^\star)$, while the reference is an action marginal $q \in \operatorname{int}\Delta(\mathcal{A})$. Define

$$\mathrm{AIR}_{q,\gamma}(p,\nu) := \mathbb{E}_{(M,A^\star)\sim\nu} \Delta_{M,A^\star}(p) - \gamma \mathbb{E}_{a\sim p} I_\nu(A^\star; O \mid a) - \gamma D_{\mathrm{KL}}(q_\nu \| q), \tag{21}$$

where q_ν is the $A^\star$-marginal of ν. Equivalently,

$$\mathrm{AIR}_{q,\gamma}(p,\nu) = \mathbb{E}_{(M,A^\star),a,O} \left[\Delta_{M,A^\star}(\delta_a) - \gamma \log \frac{\nu(A^\star \mid a, O)}{q(A^\star)} \right].$$

Lemma 4.5 (AIR gradient bracket)**.** *Assume finite spaces, or dominated spaces where the displayed Gateaux derivatives exist. Up to an additive constant on the simplex,*

$$\frac{\partial}{\partial \nu(M, a^\star)} \mathrm{AIR}_{q,\gamma}(p,\nu) = \Delta_{M,a^\star}(p) - \gamma \mathbb{E}_{a\sim p, O\sim P_{M,a}} \log \frac{\nu(A^\star = a^\star \mid a, O)}{q(a^\star)}. \tag{22}$$

Consequently, for every fixed pair $(M, a^\star)$,

$$\begin{aligned} B^{\mathrm{AIR}}_{q,\gamma}(p,\nu; M, a^\star) &:= \mathrm{AIR}_{q,\gamma}(p,\nu) + \left\langle \nabla_\nu \mathrm{AIR}_{q,\gamma}(p,\nu), \delta_{(M,a^\star)} - \nu \right\rangle \\ &= \Delta_{M,a^\star}(p) - \gamma \mathbb{E}_{a\sim p, O\sim P_{M,a}} \log \frac{\nu(A^\star = a^\star \mid a, O)}{q(a^\star)}. \end{aligned} \tag{23}$$

Proof. For fixed a, set

$$w_{a^\star,o} = \sum_M \nu(M, a^\star) P_{M,a}(o), \qquad w_o = \sum_{a^\star} w_{a^\star,o}.$$

Then

$$I_\nu(A^\star; O \mid a) + D_{\mathrm{KL}}(q_\nu \| q) = \sum_{a^\star, o} w_{a^\star, o} \log \frac{w_{a^\star, o}}{w_o q(a^\star)}.$$

Differentiating with respect to $\nu(M, a^\star)$ gives

$$\mathbb{E}_{O \sim P_{M,a}} \log \frac{\nu(A^\star = a^\star \mid a, O)}{q(a^\star)},$$

because the $+1$ terms from differentiating $w_{a^\star,o} \log w_{a^\star,o}$ and $w_o \log w_o$ cancel. Adding the derivative of the linear regret term proves (22). Pairing the gradient with $\delta_{(M,a^\star)} - \nu$ cancels the posterior-averaged terms and gives (23). □

Theorem 4.6 (Exact AIR regret identity)**.** *Let an algorithm generate decision laws p_t, pair beliefs ν_t, and reference action beliefs $q_t \in \operatorname{int} \Delta(\mathcal{A})$. After action a_t and observation O_t, set*

$$q_{t+1}(\cdot) = \nu_t(A^\star \in \cdot \mid a_t, O_t).$$

Fix a model $M^\star$ and a comparator $a^\star$ with $q_t(a^\star) > 0$ for all relevant t; in the standard stochastic case, take $a^\star \in \arg\max_a r_{M^\star}(a)$. Let

$$\mathrm{Reg}_T(\mathsf{Alg}, M^\star, a^\star) := \mathbb{E} \sum_{t=1}^{T} \left(r_{M^\star}(a^\star) - r_{M^\star}(A_t)\right).$$

Then, for every $\gamma > 0$,

$$\mathrm{Reg}_T(\mathsf{Alg}, M^\star, a^\star) = \gamma \mathbb{E} \log \frac{q_{T+1}(a^\star)}{q_1(a^\star)} + \mathbb{E} \sum_{t=1}^{T} B^{\mathrm{AIR}}_{q_t,\gamma}(p_t, \nu_t; M^\star, a^\star). \tag{24}$$

Remark 4.7 (Adversarial model sequence)**.** *Since Lemma 4.5 holds pointwise for every model-action pair, and does not require the true model to be stochastic, Theorem 4.6 extends directly to an adversarial model sequence $\mathbf{M} = (M_1, \ldots, M_T) \in \mathcal{M}^T$. In this case, the comparator is the best-in-hindsight action $a^\star$, and the regret identity remains exact. We omit this extension only to avoid additional notation.*

Proof. Condition on H_{t-1}. By the definition of the action-marginal posterior,

$$\mathbb{E}\left[\log \frac{q_{t+1}(a^\star)}{q_t(a^\star)} \middle| H_{t-1}\right] = \mathbb{E}_{a \sim p_t, O \sim P_{M^\star,a}} \log \frac{\nu_t(A^\star = a^\star \mid a, O)}{q_t(a^\star)}.$$

Combining this equality with (23) gives a one-step equality between regret, bracket, and posterior-ratio increment. Taking expectations and summing over time telescopes the logarithm. □

MAIR. For MAIR the belief is $\mu \in \Delta(\mathcal{M})$ and the reference is $\rho \in \operatorname{int} \Delta(\mathcal{M})$. Write $\mu_{O|a} = \int P_{M,a} \mu(dM)$ and

$$\Delta_M(p) = r_M(a^\star_M) - \mathbb{E}_{a \sim p} r_M(a), \qquad a^\star_M \in \arg\max_a r_M(a).$$

Define

$$\mathrm{MAIR}_{\rho,\gamma}(p, \mu) := \mathbb{E}_{M \sim \mu} \Delta_M(p) - \gamma \mathbb{E}_{a \sim p} I_\mu(M; O \mid a) - \gamma D_{\mathrm{KL}}(\mu \| \rho). \tag{25}$$

The same differentiation gives

$$\begin{aligned} B^{\mathrm{MAIR}}_{\rho,\gamma}(p,\mu;M^\star) &:= \mathrm{MAIR}_{\rho,\gamma}(p,\mu) + \langle \nabla_\mu \mathrm{MAIR}_{\rho,\gamma}(p,\mu), \delta_{M^\star} - \mu\rangle \\ &= \Delta_{M^\star}(p) - \gamma \mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{M^\star,a} \| \mu_{O|a}) - \gamma \log \frac{\mu(M^\star)}{\rho(M^\star)}. \end{aligned} \tag{26}$$

If $\rho_{t+1}(\cdot) = \mu_t(\cdot \mid a_t, O_t)$, then

$$\mathrm{Reg}_T(\mathsf{Alg}, M^\star) = \gamma \mathbb{E} \log \frac{\rho_{T+1}(M^\star)}{\rho_1(M^\star)} + \mathbb{E} \sum_{t=1}^{T} B^{\mathrm{MAIR}}_{\rho_t,\gamma}(p_t, \mu_t; M^\star). \tag{27}$$

The proof is the same posterior-ratio telescope as for AIR. Thus AIR and MAIR are one indexed identity with different choices of Y.

Remark 4.8 (Danskin, Nash equilibrium, and why the bracket is not accidental)**.** *The direct proofs above are intentionally short. The original AIR proof of Xu and Zeevi (2025, Section 5) gives the variational reason behind the same equations. Introduce an auxiliary prediction rule $Q : (a, o) \mapsto \Delta(\mathcal{A})$ and the posterior-scoring objective*

$$\mathbb{E}_{(M,A^\star),a,O}\left[\Delta_{M,A^\star}(\delta_a) - \gamma \log \frac{Q[a,O](A^\star)}{q(A^\star)}\right].$$

For fixed ν, the logarithmic score is strongly convex in the predictive distribution and is uniquely minimized by the Bayesian posterior $Q_\nu[a,o] = \nu(A^\star \in \cdot \mid a, o)$. The optimized value is the AIR objective; the unoptimized objective is linear, hence concave, in ν. Danskin's theorem identifies the AIR gradient with the derivative at the optimizer Q_ν. This is the conceptual content of Lemma 5.3 in Xu and Zeevi (2025). The same concavity plus strong convexity also gives the Nash equilibrium statement of Lemma 5.2 by first-order/Danskin reasoning, without invoking a separate fixed-point theorem. This does not necessarily simplify the full proof, because compactness, differentiability, and boundary conditions still have to be checked. It does explain why the gradient bracket is not an accidental cancellation: posterior scoring produces the minimizer, Danskin produces the gradient, and the regret identity telescopes the posterior reference.

The perspective developed in Xu and Zeevi (2025) is that frequentist sequential learning can be organized around algorithmic beliefs μ, interpreted as predictive laws, and decision rules induced by reference or posterior beliefs ρ, either through posterior sampling or through robust minimax optimization. We use MAIR to illustrate how a single functional captures both the confidence multiplier in UCB-type algorithms and the decision-estimation coefficient in one-round minimax rules.

Lemma 4.9 (MAIR gradient bracket and three canonical specializations)**.** *Assume the observation laws are dominated and the displayed derivatives exist. For fixed p, ρ, μ, up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial \mu(M)} \mathrm{MAIR}_{\rho,\gamma}(p,\mu) = \Delta_M(p) - \gamma \log \frac{\mu(M)}{\rho(M)} - \gamma \mathbb{E}_{\pi\sim p} D_{\mathrm{KL}}(P_{M,\pi} \| \mu_{o|\pi}).$$

Consequently, for every $M^\star$,

$$\begin{aligned} &\mathrm{MAIR}_{\rho,\gamma}(p,\mu) + \langle \nabla_\mu \mathrm{MAIR}_{\rho,\gamma}(p,\mu), \delta_{M^\star} - \mu\rangle \\ &\quad = \Delta_{M^\star}(p) - \gamma \mathbb{E}_{\pi\sim p} D_{\mathrm{KL}}(P_{M^\star,\pi} \| \mu_{o|\pi}) - \gamma \log \frac{\mu(M^\star)}{\rho(M^\star)}. \end{aligned} \tag{28}$$

This bracket has three important specializations.

1. *If $\mu = \rho$, the logarithmic correction vanishes and the bracket becomes the KL-DEC offset*

$$\Delta_{M^\star}(p) - \gamma \mathbb{E}_{\pi\sim p} D_{\mathrm{KL}}(P_{M^\star,\pi} \| \mu_{o|\pi}). \tag{29}$$

In Section 5 we will show that for GP-UCB,

$$\textit{self-normalized calibration} + \textit{UCB optimism} \quad \Rightarrow \quad \textit{MAIR gradient-error condition.}$$

2. *If $\bar{\mu} \in \arg\max_{\mu\in\Delta(\mathcal{M})} \mathrm{MAIR}_{\rho,\gamma}(p,\mu)$ is an interior maximizer, then*

$$\mathrm{MAIR}_{\rho,\gamma}(p,\bar{\mu}) + \langle \nabla_\mu \mathrm{MAIR}_{\rho,\gamma}(p,\bar{\mu}), \delta_{M^\star} - \bar{\mu}\rangle \le \mathrm{MAIR}_{\rho,\gamma}(p,\bar{\mu}). \tag{30}$$

This is the MAMS/MEBO bracket control.

3. *Let $p_{M,\pi} = dP_{M,\pi}/d\nu_\pi$ and define, for an auxiliary pair $(\bar{\pi},\bar{o})$,*

$$\frac{d\mu^{1/2}_{\bar{\pi},\bar{o}}}{d\rho}(M) := \frac{\sqrt{p_{M,\bar{\pi}}(\bar{o})}}{\int \sqrt{p_{N,\bar{\pi}}(\bar{o})}\,\rho(dN)}. \tag{31}$$

Let $h^2(P,Q) := 1 - \int \sqrt{dP\,dQ}$. If $\bar{\pi} \sim p$ and $\bar{o} \sim P_{M^\star,\bar{\pi}}$, then

$$\begin{aligned}\mathbb{E}_{\bar{\pi},\bar{o}}\Big[&\mathrm{MAIR}_{\rho,\gamma}(p,\mu^{1/2}_{\bar{\pi},\bar{o}}) + \langle \nabla_\mu \mathrm{MAIR}_{\rho,\gamma}(p,\mu^{1/2}_{\bar{\pi},\bar{o}}), \delta_{M^\star} - \mu^{1/2}_{\bar{\pi},\bar{o}}\rangle\Big]\\ &\le \Delta_{M^\star}(p) - \gamma \mathbb{E}_{\pi\sim p}\mathbb{E}_{M\sim\rho} h^2(P_{M^\star,\pi}, P_{M,\pi}). \end{aligned}\tag{32}$$

Thus the square-root posterior turns the MAIR bracket into a Hellinger DEC offset.

Proof. The derivative calculation is direct. The derivative of $\mathbb{E}_\mu \Delta_M(p)$ is $\Delta_M(p)$; the derivative of $D_{\mathrm{KL}}(\mu\|\rho)$ is $\log(\mu(M)/\rho(M)) + 1$; and differentiating the predictive mixture in the mutual-information term gives $D_{\mathrm{KL}}(P_{M,\pi}\|\mu_{o|\pi})$, up to the same simplex constant. Pairing the derivative with $\delta_{M^\star} - \mu$ cancels the averaged terms in (25) and proves (28).

The case $\mu = \rho$ is immediate from the logarithmic cancellation.

The case $\bar{\mu} \in \arg\max \mathrm{MAIR}$ follows from first-order optimality.

It remains to justify the square-root claim; this is a standard fact for the squared Hellinger distance. For fixed $\bar{\pi}$, let

$$Z_{\bar{\pi}}(\bar{o}) := \int \sqrt{p_{N,\bar{\pi}}(\bar{o})}\,\rho(dN).$$

Then

$$\log \frac{d\mu^{1/2}_{\bar{\pi},\bar{o}}}{d\rho}(M^\star) = \frac{1}{2}\log p_{M^\star,\bar{\pi}}(\bar{o}) - \log Z_{\bar{\pi}}(\bar{o}).$$

Under $\bar{o} \sim P_{M^\star,\bar{\pi}}$, Jensen's inequality gives

$$\begin{aligned}\mathbb{E}\log \frac{d\mu^{1/2}_{\bar{\pi},\bar{o}}}{d\rho}(M^\star) &= -\mathbb{E}\log \frac{Z_{\bar{\pi}}(\bar{o})}{\sqrt{p_{M^\star,\bar{\pi}}(\bar{o})}}\\ &\ge -\log \int \sqrt{p_{M^\star,\bar{\pi}}(o)} Z_{\bar{\pi}}(o) d\nu_{\bar{\pi}}(o)\\ &= -\log\big(1 - \mathbb{E}_{M\sim\rho} h^2(P_{M^\star,\bar{\pi}}, P_{M,\bar{\pi}})\big)\\ &\ge \mathbb{E}_{M\sim\rho} h^2(P_{M^\star,\bar{\pi}}, P_{M,\bar{\pi}}).\end{aligned}$$

Averaging over $\bar{\pi} \sim p$, substituting into (28), and dropping the nonnegative KL term proves (32). $\square$

5 Upper bounds: one identity and four algorithm families

We first isolate the fixed-truth log-potential telescope, because it is the pathwise algebra underneath AIR/MAIR upper bounds. Bayesian mutual information, UCB self-normalized gain, and EBO log-partition relaxations are different ways of controlling this telescope; they should not be conflated. Fix a truth $\omega^\star$, write $y^\star = \chi(\omega^\star)$, and let q_s be the maintained reference law of the index at state s. This reference law may be an exact posterior marginal, an algorithmic belief marginal, a confidence-to-belief construction, or an exponential-weights marginal. After action a and observation o, write $q^+(\cdot \mid s, a, o)$ for the next reference marginal. The fixed-truth reference log gain is

$$\mathsf{G}_\chi(s, p; \omega^\star) := \mathbb{E}_{a\sim p,\, O\sim P_{\omega^\star,s,a}} \log \frac{q^+(y^\star \mid s, a, O)}{q_s(y^\star)}. \tag{33}$$

When the reference update is exact indexed Bayes, this is $J_\chi(s, p; \omega^\star)$ from (16). When the reference is an AIR pair belief ν with marginal q, it is the AIR posterior-coordinate log ratio

$$\mathbb{E}_{a\sim p, O\sim P_{\omega^\star,a}} \log \frac{\nu(Y = y^\star \mid a, O)}{q(y^\star)}.$$

For any predictable continuation potential V_t on states and any coefficient $\gamma > 0$, define the specialized fixed-truth logarithmic potential

$$\Phi_t^{V,\gamma}(s; y^\star) := V_t(s) + \gamma \log \frac{1}{q_s(y^\star)}. \tag{34}$$

The corresponding AIR/MAIR Bellman bracket is

$$\mathfrak{B}_\chi^{V,\gamma}(s, p; \omega^\star) := \ell_{\omega^\star}(s, p) + \mathbb{E}_{\omega^\star}[V_{t+1}(S^+) \mid s, p] - V_t(s) - \gamma \mathsf{G}_\chi(s, p; \omega^\star), \tag{35}$$

where $S^+ = \tau(s, A, O)$ with $A \sim p$ and $O \sim P_{\omega^\star,s,A}$. The identity behind the whole upper section is the telescope

$$\begin{aligned}\mathbb{E}_{\omega^\star} L_{\omega^\star}(H_T) = V_1(s_1) - \mathbb{E}_{\omega^\star} V_{T+1}(S_{T+1}) + \gamma \log \frac{1}{q_1(y^\star)} - \gamma \mathbb{E}_{\omega^\star} \log \frac{1}{q_{T+1}(y^\star)} \\ + \mathbb{E}_{\omega^\star} \sum_{t=1}^{T} \mathfrak{B}_\chi^{V,\gamma}(S_t, p_t; \omega^\star).\end{aligned} \tag{36}$$

Thus an upper bound is a Bellman supersolution for this log-potential bracket. The four algorithm families below differ only in how they make the bracket small: the exact log-penalized Bellman program optimizes it dynamically; UCB certifies it by calibration and optimism; E2D drops or bounds the continuation to obtain a one-step offset; and AMS/EBO optimizes a robust KL-dual belief relaxation whose first-order condition certifies it.

Theorem 5.1 (Generic fixed-truth indexed AIR/MAIR upper certificate)**.** *Fix $\omega^\star$ and $y^\star = \chi(\omega^\star)$. Assume $q_1(y^\star) > 0$ and that the reference update remains positive on $y^\star$ along the analyzed trajectory. If an algorithm chooses p_t so that, for every reached state,*

$$\mathfrak{B}_\chi^{V,\gamma}(S_t, p_t; \omega^\star) \le \varepsilon_t,$$

then

$$\mathbb{E}_{\omega^\star}^{\mathsf{Alg}} L_{\omega^\star}(H_T) \le V_1(s_1) + \gamma \log \frac{1}{q_1(y^\star)} - \mathbb{E}_{\omega^\star}^{\mathsf{Alg}} V_{T+1}(S_{T+1}) - \gamma \mathbb{E}_{\omega^\star}^{\mathsf{Alg}} \log \frac{1}{q_{T+1}(y^\star)} + \mathbb{E}_{\omega^\star}^{\mathsf{Alg}} \sum_{t=1}^{T} \varepsilon_t. \tag{37}$$

Proof. Taking the conditional expectation of $\gamma \log(1/q_{t+1}(y^\star)) - \gamma \log(1/q_t(y^\star))$ under $P_{\omega^\star,s,a}$ gives $-\gamma \mathsf{G}_\chi(s, p; \omega^\star)$. Substituting the bracket inequality into the telescoping identity (36) proves (37). □

It is useful to keep one concrete state specialization in mind. A fixed estimation procedure maps histories to algorithmic beliefs ν_t over environments, a decision procedure maps the retained state to p_t, and the index marginal is

$$q_t = \chi_\# \nu_t. \tag{38}$$

In a well-specified Bayes analysis, ν_t is the exact posterior. In a frequentist analysis, ν_t may be only an algorithmic belief or confidence object; then a calibration event must put the fixed truth into the comparison set used by the robust Bellman program below. Posterior averaging of the coordinate potential gives

$$\mathbb{E}_{Y \sim \nu_\chi} \gamma \log \frac{1}{q_s(Y)} = \gamma H(\nu_\chi) + \gamma D_{\mathrm{KL}}(\nu_\chi \| q_s), \tag{39}$$

and for a finite index set the same logarithmic scoring rule has the KL-dual log-partition form

$$\Psi_\gamma(q, z) := \gamma \log \sum_{y \in \mathcal{Y}} q(y) e^{z_y/\gamma} = \sup_{r \in \Delta(\mathcal{Y})} \left\{ \langle r, z \rangle - \gamma D_{\mathrm{KL}}(r \| q) \right\}. \tag{40}$$

The coordinate log loss, its posterior average, and the dual log partition are three representations of the same KL/logarithmic scoring rule. They are not interchangeable in a proof: fixed-truth guarantees use (34), posterior-averaged Bayes identities use (39), and AMS/EBO relaxations use (40).

In the model-index specialization, write the posterior/reference belief as μ_t. If the maintained reference update is exact Bayes, then $\mathsf{G}_\chi = J_\chi$ and the fixed-truth MAIR information increment is

$$J_t^{\mathrm{MAIR}}(p; \omega^\star) := \mathbb{E}_{a \sim p} D_{\mathrm{KL}}(P_{\omega^\star,t}(\cdot \mid S_t, a) \,\|\, P_{\mu_t,t}(\cdot \mid S_t, a)). \tag{41}$$

The action-index AIR version is identical with the index $Y = A^\star$, the reference marginal q_t, and the AIR posterior coordinate $q^+(a^\star \mid a, o)$ in place of the model posterior.

5.1 Information-potential Bellman programming

The exact frequentist upper algorithm is a log-penalized Bellman program on calibrated state/index representations. Let $\mathfrak{C}_t(s)$ be the set of environments that the algorithmic state regards as admissible at state s. The set may be a confidence set, a localized posterior support, or a robust belief support. For $\gamma > 0$, define

$$W^\gamma_{T+1}(s) = 0, \qquad W^\gamma_t(s) := \inf_{p \in \Delta(\mathcal{A}_t(s))} \sup_{\omega \in \mathfrak{C}_t(s)} \left\{ \ell_\omega(s, p) + \mathbb{E}_\omega[W^\gamma_{t+1}(S^+) \mid s, p] - \gamma \mathsf{G}_\chi(s, p; \omega) \right\}. \tag{42}$$

A policy attaining the infimum in (42) is the information-penalized Bellman dynamic-programming algorithm. This construction builds on the principle of admissible relaxations (Rakhlin et al., 2012), which organize learning through dynamic programming over loss states, and is particularly close to the partial-information relaxation viewpoint of Rakhlin and Sridharan (2016), where estimator-like state variables and potential terms quantify uncertainty. It can also be viewed as a sequential analogue of classical minimum-complexity, information-risk, and Gibbs-type estimation in fixed experiments (Barron and Cover, 1991; Yang and Barron, 1999; Zhang, 2006). The key new feature is that a generic logarithmic mass penalty is propagated through a controlled Bellman recursion, rather than applied only once to a terminal estimator, and the recursion is carried by a Bellman-sufficient state together with an explicit information index.

Theorem 5.2 (Frequentist log-penalized Bellman upper bound)**.** *Fix $\Omega_0 \subseteq \Omega$ and $\gamma > 0$. Suppose the reference update and the comparison sets $\mathfrak{C}_t$ are calibrated for Ω_0 in the following sense: for every $\omega^\star \in \Omega_0$, on the calibration event $\mathcal{E}$, $\omega^\star \in \mathfrak{C}_t(S_t)$ whenever round t is reached. Suppose the algorithm chooses p_t so that*

$$\sup_{\omega \in \mathfrak{C}_t(S_t)} \left\{ \ell_\omega(S_t, p_t) + \mathbb{E}_\omega[W^\gamma_{t+1}(S^+) \mid S_t, p_t] - \gamma \mathsf{G}_\chi(S_t, p_t; \omega) \right\} \le W^\gamma_t(S_t) + \varepsilon_t.$$

Then, for every fixed truth $\omega^\star \in \Omega_0$ with $y^\star = \chi(\omega^\star)$ and $q_1(y^\star) > 0$, provided the reference update remains positive on $y^\star$ on the calibrated trajectory,

$$\mathbb{E}^{\mathsf{Alg}}_{\omega^\star} L_{\omega^\star}(H_T) \le W^\gamma_1(s_1) + \gamma \log \frac{1}{q_1(y^\star)} + \mathbb{E}^{\mathsf{Alg}}_{\omega^\star} \sum_{t=1}^{T} \varepsilon_t + \mathrm{cal}_T(\omega^\star), \tag{43}$$

provided the index is finite or countable, where

$$\mathrm{cal}_T(\omega^\star) := \mathbb{E}_{\omega^\star}\big[L_{\omega^\star}(H_T)^+ \mathbb{1}\{\mathcal{E}^c\}\big].$$

If calibration holds surely, then $\mathrm{cal}_T(\omega^\star) = 0$; if the positive part of the cumulative loss is bounded by B_T, then $\mathrm{cal}_T(\omega^\star) \le B_T \mathbb{P}_{\omega^\star}(\mathcal{E}^c)$. The common per-round bound $0 \le \ell_t \le L$ is only the special case $B_T = LT$.

Proof. On the calibration event, the fixed truth is one of the environments in the robust supremum. Therefore the displayed approximate minimization implies

$$\ell_{\omega^\star}(S_t, p_t) + \mathbb{E}_{\omega^\star}[W^\gamma_{t+1}(S^+) \mid S_t, p_t] - W^\gamma_t(S_t) - \gamma \mathsf{G}_\chi(S_t, p_t; \omega^\star) \le \varepsilon_t.$$

Apply Theorem 5.1 with $V_t = W^\gamma_t$. Since $W^\gamma_{T+1} = 0$ and the terminal index log loss is nonnegative for a finite or countable index, the result follows on $\mathcal{E}$. Splitting the expectation according to $\mathcal{E}$ and $\mathcal{E}^c$ and upper-bounding the failure contribution by the positive part gives the calibration term. □

The posterior-averaged Bayesian recursion is a specialization rather than the primitive fixed-truth statement. If the robust comparison set is replaced by the current posterior law and one averages the coordinate identity over $Y \sim q_s$, then

$$\mathbb{E}[H_\chi(S_{t+1}) \mid s, p] - H_\chi(s) = -\mathcal{I}_\chi(s, p).$$

For a fixed multiplier $\gamma > 0$, define the Bayes information-regularized value

$$U^\gamma_{T+1} \equiv 0, \qquad U^\gamma_t(s) := \inf_{p \in \Delta(\mathcal{A}_t(s))} \left\{ \ell(s, p) + \mathbb{E}[U^\gamma_{t+1}(S_{t+1}) \mid s, p] - \gamma \mathcal{I}_\chi(s, p) \right\}. \tag{44}$$

Equivalently,

$$\Phi^\gamma_t(s) := U^\gamma_t(s) + \gamma H_\chi(s) \tag{45}$$

solves the admissible-relaxation recursion

$$\Phi^\gamma_t(s) = \inf_{p \in \Delta(\mathcal{A}_t(s))} \left\{ \ell(s, p) + \mathbb{E}[\Phi^\gamma_{t+1}(S_{t+1}) \mid s, p] \right\}, \qquad \Phi^\gamma_{T+1}(s) = \gamma H_\chi(s). \tag{46}$$

Theorem 5.3 (Posterior-averaged information-potential upper bound)**.** *Assume the indexed Bellman representation is exact under the Bayesian mixture law and $H_\mu(Y) < \infty$. If a policy chooses p_t satisfying*

$$\ell(S_t, p_t) + \mathbb{E}[U^\gamma_{t+1}(S_{t+1}) \mid S_t, p_t] - \gamma \mathcal{I}_\chi(S_t, p_t) \le U^\gamma_t(S_t) + \varepsilon_t,$$

then

$$\mathbb{E}L_\Omega(H_T) \le U_1^\gamma(s_1) + \gamma I_\mu(Y; H_T) + \sum_{t=1}^{T} \varepsilon_t \le U_1^\gamma(s_1) + \gamma H_\mu(Y) + \sum_{t=1}^{T} \varepsilon_t. \tag{47}$$

This is the posterior average of the fixed-truth coordinate telescope, not a pointwise replacement for Theorem 5.2.

Proof. Rearrange the displayed inequality and telescope U_t^γ. Proposition 4.2 identifies $\mathbb{E}\sum_t \mathcal{I}_\chi(S_t, p_t)$ with $I_\mu(Y; H_T)$, and $I_\mu(Y; H_T) \le H_\mu(Y)$ gives the final bound. The equivalence with (46) follows from the posterior entropy-drop identity. □

The minimax information-risk sandwich and the entropy/regret comparison are stated in Section 3. The rest of this upper-bound section develops the upper side of that sandwich: the fixed-truth coordinate telescope, the exact log-penalized Bellman program, and the main tractable relaxations or certificates.

5.2 UCB families: calibration plus optimism

An upper-confidence-bound (UCB) algorithm maintains a prediction and an uncertainty scale, builds a confidence band from them, and chooses an action that is optimistic within that band (Auer et al., 2002; Abbasi-Yadkori et al., 2011; Srinivas et al., 2010). We use the following notation throughout. The unmultiplied uncertainty scale is denoted by $\sigma_t(a)$ in the generic discussion and by $s_t(x)$ in the GP specialization below. The confidence half-width is

$$w_t(a) = \beta_t \sigma_t(a),$$

where β_t is the confidence multiplier. Calibration says that the fixed truth is contained in the band of radius $w_t(a)$; optimism says that the selected action maximizes the upper envelope. Together, in reward-maximization problems, these two facts give

$$r_{\omega^\star}(s, a^\star) \le m_t(a^\star) + w_t(a^\star) \le m_t(a_t) + w_t(a_t) \le r_{\omega^\star}(s, a_t) + 2w_t(a_t),$$

and hence the instantaneous regret is at most $2\beta_t\sigma_t(a_t)$. An elliptical-potential/log-determinant lemma then controls the realized sum of $\sigma_t^2(a_t)$. Thus UCB does not solve the AIR/MAIR bracket optimization directly; it certifies the relevant fixed-truth bracket after the optimistic action has been chosen.

Lemma 5.4 (Calibration and optimism imply bracket control)**.** *Fix a truth $\omega^\star$ and condition on an event $\mathcal{E}$ on which the algorithm's state is calibrated for that truth. Suppose that, for every reached round t, the action a_t chosen by the algorithm and a predictable uncertainty scale $\sigma_t(a) \ge 0$ satisfy*

$$\ell_{\omega^\star}(S_t, a_t) \le c_{\mathrm{opt}}\, \beta_t \sigma_t(a_t), \tag{48}$$

where c_{opt} is a numerical optimism constant; in the usual two-sided reward-confidence proof above, $c_{\mathrm{opt}} = 2$. Then for every $\gamma > 0$,

$$\ell_{\omega^\star}(S_t, a_t) - \gamma\sigma_t^2(a_t) \le \frac{c_{\mathrm{opt}}^2 \beta_t^2}{4\gamma}. \tag{49}$$

Consequently, for the deterministic-action fixed-truth bracket in (18),

$$B_{\chi,\gamma}(S_t, \delta_{a_t}; \omega^\star) \le \frac{c_{\mathrm{opt}}^2 \beta_t^2}{4\gamma} + \gamma \left(\sigma_t^2(a_t) - J_\chi(S_t, \delta_{a_t}; \omega^\star)\right), \tag{50}$$

where the fixed-truth index log gain is defined in (16)*. If, in addition, the log-determinant/elliptical-potential argument gives a realized bound*

$$\sum_{t=1}^{T} \sigma_t^2(a_t) \le c_{\mathrm{sn}} G_T,$$

for an information telescope G_T*, then summing* (50) *reduces UCB regret control to the comparison between* G_T *and the fixed-truth log-gain telescope.*

Proof. The optimism/calibration hypothesis (48) and Young's inequality give

$$c_{\mathrm{opt}} \beta_t \sigma_t(a_t) \le \frac{c_{\mathrm{opt}}^2 \beta_t^2}{4\gamma} + \gamma \sigma_t^2(a_t).$$

This proves (49). Subtracting $\gamma J_\chi(S_t, \delta_{a_t}; \omega^\star)$ from both sides and using the definition of $B_{\chi,\gamma}$ proves (50). The final sentence is the result of summing the displayed inequality over time. □

The correction $\sigma_t^2(a_t) - J_\chi(S_t, \delta_{a_t}; \omega^\star)$ is not a second estimation cost. It is the algebraic difference between the tractable self-normalized variance proxy and the fixed-truth posterior log gain appearing in the bracket. In linear-UCB and GP–UCB proofs, the log-determinant or elliptical-potential argument is precisely the certificate that the realized sum of $\sigma_t^2(a_t)$ is controlled by an information telescope. The confidence multiplier β_t is separate: it is the price of converting calibrated uncertainty into regret through optimism. Therefore

$$\text{self-normalized calibration} + \text{optimism} \quad \Longrightarrow \quad \text{fixed-truth AIR/MAIR bracket control.}$$

In this organizing perspective, the round-heterogeneous summation of unmultiplied uncertainty scales, usually controlled by an elliptical-potential or log-determinant argument, plays the role of a realized estimation-complexity telescope. The confidence multiplier enters instead through the regret-to-uncertainty conversion, and hence through the coefficient in the fixed-truth bracket. This separates two mechanisms that are often blurred in informal UCB discussions: posterior or confidence widths quantify the "estimation complexity" of the underlying parameter, as explained by Lattimore (2023), while optimism and calibration determine how that geometry pays for regret. The resulting unification reveals a common posterior-telescoping structure while keeping distinct the estimation telescope and the coefficient mechanisms behind UCB, E2D, and AMS/EBO.

5.3 E2D: robust one-step offset optimization

The estimation-to-decision (E2D) algorithms (Foster et al., 2021) choose p by solving a one-round robust DEC offset at the current time-state pair. In model-index notation the schematic KL form is

$$\inf_{p \in \Delta(\mathcal{A}_t(s))} \sup_{\omega \in \mathfrak{C}_t(s)} \left\{ \ell_\omega(s,p) - \gamma \mathbb{E}_{a\sim p} D_{\mathrm{KL}}(P_{\omega,s,a} \| P_{\rho,s,a}) \right\}, \tag{51}$$

where ρ is a reference belief or reference model and $\mathfrak{C}_t(s)$ is a localized comparison set that contains the fixed truth on the calibration event. The notation is intentionally the same as in the log-penalized Bellman program: E2D uses the current feasible action set and the current localized comparison set, but it replaces the dynamic continuation by a one-step separation penalty. The same one-round minimax principle applies to the original Hellinger form and constrained variants (Foster et al., 2023).

What E2D keeps is the immediate regret and a one-step statistical-separation penalty. What it drops, freezes, or upper bounds is the Bellman continuation term

$$\mathbb{E}_\omega[V_{t+1}(S^+) \mid s, p] - V_t(s),$$

together with the exact coordinate reference log gain G_χ in (35). Thus E2D is not an optimizer of the full Bellman potential unless one proves that the omitted continuation is dominated by the displayed one-step penalty. In an exact Bayesian calculation, posterior averaging can turn the KL penalty into a one-step mutual-information term. In a fixed-truth or frequentist calculation, the corresponding statement is the reference posterior-ratio identity for G_χ, or its exact-Bayes specialization J_χ, plus whatever calibration or localization is needed to compare $P_{\rho,s,a}$ with the reference predictive law in that identity. Without this compatibility, one may pay for estimation difficulty once through the one-step DEC and again through a separate log-determinant or posterior-ratio telescope. The safe interpretation is therefore: E2D is a one-step relaxation, re-solved at each state, of the fixed-truth log-posterior Bellman bracket.

5.4 AMS/EBO: robust convex belief optimization

The Exploration by Optimization (EBO) algorithm (Lattimore and György, 2021; Foster et al., 2022) and the Adaptive Minimax Sampling (AMS) algorithm (Xu and Zeevi, 2025; Liu et al., 2025, 2026) should be understood as tractable relaxations of the robust log-penalized Bellman program. AMS may be viewed as a constructive implementation of the EBO principle, replacing abstract functional-estimator optimization with an executable sampling rule. We therefore focus on the AMS rule here. Work with an action or policy index $Y = \chi(\Omega)$. At state s, let $q \in \operatorname{int}\Delta(\mathcal{Y})$ be the current reference marginal and let $\mathfrak{B}_t(s)$ be a calibrated convex set of pair beliefs ν on (ω, y), supported on $y = \chi(\omega)$. For each ν, write ν_χ for its index marginal and define

$$\ell_\nu(s,p) = \mathbb{E}_{(\omega,y)\sim\nu}\ell_\omega(s,p), \qquad \mathcal{I}_\chi(s,p;\nu) = \mathbb{E}_{a\sim p} I_\nu(Y; O \mid s, a).$$

With coefficient $\gamma > 0$, the one-step robust AIR/EBO objective is

$$\mathfrak{A}_{q,\gamma}(s,p,\nu) := \ell_\nu(s,p) - \gamma\mathcal{I}_\chi(s,p;\nu) - \gamma D_{\mathrm{KL}}(\nu_\chi \| q). \tag{52}$$

The robust maximization over beliefs is essential:

$$p_t \in \operatorname*{arg\,min}_{p\in\Delta(\mathcal{A}_t(s))} \max_{\nu\in\mathfrak{B}_t(s)} \mathfrak{A}_{q_t,\gamma}(s,p,\nu), \qquad \bar{\nu}_t \in \operatorname*{arg\,max}_{\nu\in\mathfrak{B}_t(s)} \mathfrak{A}_{q_t,\gamma}(s,p_t,\nu). \tag{53}$$

Without the inner $\max_\nu$, the display is only a posterior AIR calculation for a chosen belief, not a robust AMS/EBO rule.

The convex-analytic interpretation is the following. The fixed-truth proof uses the coordinate log loss $\gamma \log(1/q_t(y^\star))$. Averaging this coordinate under a candidate belief gives $\gamma H(\nu_\chi) + \gamma D_{\mathrm{KL}}(\nu_\chi \| q_t)$. When the next reference marginal is the posterior coordinate induced by ν, the expected change of this averaged potential is $-\gamma\mathcal{I}_\chi(s,p;\nu) - \gamma D_{\mathrm{KL}}(\nu_\chi \| q_t)$. Equivalently, optimizing over possible next index laws yields the log-partition dual (40). Hence the EBO potential is the KL-dual log partition, while the fixed-truth AIR certificate is the coordinate posterior log loss; they coincide only through KL duality and the chosen update, not as identical scalar functions.

For a genuine convex belief optimization one must verify the convexity/concavity assumptions used by the saddle argument. In finite exponential-weights or admissible-relaxation settings this follows from the log-partition dual. In general model classes, mutual information need not be concave in an arbitrary belief when both the index marginal and the observation kernels vary. One must either verify concavity of $\nu \mapsto \mathfrak{A}_{q,\gamma}(s,p,\nu)$ on $\mathfrak{B}_t(s)$, or replace it by a certified concave upper relaxation.

Lemma 5.5 (AMS/EBO saddle certifies the AIR fixed-truth bracket)**.** *Assume the finite or dominated differentiability setting of Lemma 4.5. Fix $(\omega^\star, y^\star)$ with $y^\star = \chi(\omega^\star)$. Suppose $\nu \mapsto \mathfrak{A}_{q_t,\gamma}(s, p_t, \nu)$ is concave on the convex set $\mathfrak{B}_t(s)$, $\bar{\nu}_t$ maximizes it as in (53), and the comparison direction $\delta_{(\omega^\star,y^\star)} - \bar{\nu}_t$ is admissible for $\mathfrak{B}_t(s)$; any failure of this containment is charged as a calibration error. If*

$$\max_{\nu \in \mathfrak{B}_t(s)} \mathfrak{A}_{q_t,\gamma}(s, p_t, \nu) \le \varepsilon_t,$$

then the fixed-truth AIR bracket with coefficient γ satisfies

$$\ell_{\omega^\star}(s, p_t) - \gamma \mathbb{E}_{a \sim p_t, O \sim P_{\omega^\star,s,a}} \log \frac{\bar{\nu}_t(Y = y^\star \mid s, a, O)}{q_t(y^\star)} \le \varepsilon_t. \tag{54}$$

If the reference update is $q_{t+1}(\cdot) = \bar{\nu}_t(Y \in \cdot \mid s, A_t, O_t)$ and the calibration assumptions hold along the trajectory, then Theorem 5.1 with $V_t \equiv 0$ gives

$$\mathbb{E}_{\omega^\star} \operatorname{Reg}_T \le \gamma \log \frac{1}{q_1(y^\star)} + \sum_{t=1}^{T} \varepsilon_t + \mathrm{cal}_T \tag{55}$$

for finite or countable index sets. Averaging this coordinate bound under a Bayesian prior replaces $\log(1/q_1(Y))$ *by the corresponding cross-entropy or by $H_\mu(Y)$ when $q_1 = \mu_\chi$.*

Proof. The AIR gradient calculation in Lemma 4.5, identifies the fixed-truth bracket as the first-order value of $\mathfrak{A}_{q_t,\gamma}$ in the direction $\delta_{(\omega^\star,y^\star)} - \bar{\nu}_t$. Concavity and optimality of $\bar{\nu}_t$ make that first-order correction nonpositive for every admissible comparison direction. Hence the bracket is no larger than $\mathfrak{A}_{q_t,\gamma}(s, p_t, \bar{\nu}_t)$, which is at most the robust value and therefore at most ε_t. The regret bound is the fixed-truth log-potential telescope. □

A dynamic AMS/EBO relaxation is obtained by adding a continuation term before the robust maximization:

$$\mathfrak{A}^V_{t,q,\gamma}(s, p, \nu) = \ell_\nu(s, p) + \mathbb{E}_\nu V_{t+1}(S^+) - V_t(s) - \gamma \mathcal{I}_\chi(s, p; \nu) - \gamma D_{\mathrm{KL}}(\nu_\chi \| q).$$

This is a valid relaxation of the robust Bellman recursion (42) only after the same first-order or direct-bracket certificate is proved. The standard one-step EBO display is the case $V_t \equiv 0$; the dynamic version is an admissible relaxation of the central log-penalized Bellman program, not an automatic consequence of the one-step objective.

Remark 5.6 (The one identity and the four families)**.** *The identity is the fixed-truth log-potential telescope (36). The exact information-risk Bellman program is (42). Posterior-averaged Bayes analysis is obtained by averaging the same coordinate identity, giving (44). UCB certifies the bracket through confidence and optimism. E2D replaces the continuation by a one-step separation penalty. AMS/EBO optimizes a KL-dual robust belief relaxation, with the required* $\max_\nu$ *appearing explicitly. Thus the common object is not a universal one-step coefficient but the logarithmic Bellman bracket on a sufficient state and a chosen index.*

6 Lower bounds: reference histories and quantile indices

6.1 Ghost probability and true good probability

Fix an average-regret threshold $r \ge 0$. The posterior-reference ghost-good probability is

$$p_r^{\mathsf{Alg}}(\mu, \chi) := \mathbb{P}_{Y \sim \mu_\chi,\, H'_T \sim \bar{\mathbb{P}}_\mu^{\mathsf{Alg}}} \left(\bar{L}_\chi(Y, H'_T) \le r \right), \tag{56}$$

where Y and H'_T are independent under the ghost law and μ_χ is the marginal law of Y. The true good probability is

$$q_r^{\mathsf{Alg}}(\mu, \chi) := \mathbb{P}_{Y,H_T}\left(\bar{L}_\chi(Y, H_T) \le r\right), \tag{57}$$

where (Y, H_T) are generated by $\Omega \sim \mu$ and $H_T \sim \mathbb{P}_\Omega^{\mathsf{Alg}}$.

The ghost probability is algorithm-dependent. That dependence is intentional: it preserves the reference trajectory geometry instead of replacing it with a worst-case static small ball.

6.2 Reference-history quantile theorem

Let

$$\mathrm{kl}(q, p) = q \log \frac{q}{p} + (1 - q) \log \frac{1 - q}{1 - p}$$

be binary KL divergence. For $p \in [0, 1]$ and $c \ge 0$, define

$$\psi(p, c) := \sup\{q \in [0, 1] : \mathrm{kl}(q, p) \le c\}.$$

Theorem 6.1 (Reference-history quantile indexed-information lower bound)**.** *For every prior μ, index map χ, threshold $r \ge 0$, and algorithm Alg, assume that the posterior-reference process satisfies the exact posterior-indexed lift in the sense of Definition 2.2, and assume that the cumulative indexed loss satisfies $L_\chi(Y, H_T) \ge 0$ almost surely under the true mixture law,*

$$\mathrm{kl}\big(q_r^{\mathsf{Alg}}(\mu, \chi), p_r^{\mathsf{Alg}}(\mu, \chi)\big) \le C_\chi^{\mathsf{Alg}}(\mu) = T\bar{C}_\chi^{\mathsf{Alg}}(\mu). \tag{58}$$

Consequently,

$$\mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) \ge Tr\left[1 - \psi\big(p_r^{\mathsf{Alg}}(\mu, \chi), T\bar{C}_\chi^{\mathsf{Alg}}(\mu)\big)\right]. \tag{59}$$

In particular, if

$$T\bar{C}_\chi^{\mathsf{Alg}}(\mu) \le \mathrm{kl}\left(\frac{1}{2}, p_r^{\mathsf{Alg}}(\mu, \chi)\right), \tag{60}$$

then

$$\mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} \bar{L}_\Omega(H_T) \ge \frac{r}{2}, \qquad \mathbb{E}_{\Omega\sim\mu}\mathbb{E}_{\mathbb{P}_\Omega^{\mathsf{Alg}}} L_\Omega(H_T) \ge \frac{Tr}{2}. \tag{61}$$

A convenient sufficient condition is

$$T\bar{C}_\chi^{\mathsf{Alg}}(\mu) + \log 2 \le \frac{1}{2} \log \frac{1}{p_r^{\mathsf{Alg}}(\mu, \chi)}. \tag{62}$$

Proof. Consider the true joint law of (Y, H_T) and the ghost law of (Y, H'_T). Both have the same marginal law of Y, while Y and H'_T are independent under the ghost law. Hence

$$D_{\mathrm{KL}}\big(\mathcal{L}(Y, H_T) \| \mathcal{L}(Y, H'_T)\big) = I_\mu(Y; H_T).$$

Data processing applied to the indicator of the event $\{\bar{L}_\chi(Y, \cdot) \le r\}$ gives

$$\mathrm{kl}(q_r^{\mathsf{Alg}}, p_r^{\mathsf{Alg}}) \le I_\mu(Y; H_T).$$

Proposition 4.2 identifies the right-hand side with $C_\chi^{\mathsf{Alg}}(\mu) = T\bar{C}_\chi^{\mathsf{Alg}}(\mu)$, proving (58). The definition of ψ gives $q_r^{\mathsf{Alg}} \le \psi(p_r^{\mathsf{Alg}}, T\bar{C}_\chi^{\mathsf{Alg}})$. By nonnegativity of the cumulative indexed loss and by (3),

$$\begin{aligned}\mathbb{E}_{\Omega,H_T} L_\Omega(H_T) &= \mathbb{E}_{Y,H_T} L_\chi(Y, H_T)\\ &\ge Tr\,\mathbb{P}\{\bar{L}_\chi(Y, H_T) > r\}\\ &= Tr(1 - q_r^{\mathsf{Alg}}),\end{aligned}$$

which proves (59). If (60) holds, then $q_r^{\mathsf{Alg}} \le 1/2$, giving (61). Finally,

$$\mathrm{kl}\left(\frac{1}{2}, p\right) = \frac{1}{2}\log\frac{1}{4p(1-p)} \ge \frac{1}{2}\log\frac{1}{p} - \log 2,$$

so (62) implies (60). □

Relation to interactive Fano. Theorem 6.1 is an adaptation of the interactive Fano method of Chen et al. (2024) to the Bellman-sufficient representation framework. The binary-KL step is the same ghost-data argument: one compares the true experiment with a reference experiment, applies data processing to the small-regret event, and obtains a Bayes risk lower bound when the trajectory information is smaller than the logarithmic inverse ghost-good probability. The additional point here is representational. We choose the reference experiment to be the Bayesian posterior-predictive trajectory and choose an information index $Y = \chi(\Omega)$. With this choice, the trajectory information admits the exact stepwise identity through Bellman states:

$$I_\mu(Y; H_T) = \mathbb{E}_{H'_T \sim \bar{\mathbb{P}}_\mu^{\mathsf{Alg}}} \sum_{t=1}^{T} \mathcal{I}_\chi(S'_t, p'_t).$$

Thus the same reference history H'_T generates both the ghost quantile and the indexed information telescope. This is the lower-bound counterpart of the AIR/MAIR upper-bound identity. In particular, $\chi(\Omega) = \Omega$ gives the model-index MAIR form, while $\chi(\Omega) = \pi^\star(\Omega)$ gives the action-index AIR form.

Remark 6.2 (Why average notation is useful)**.** *The theorem could be written entirely with cumulative quantities. Writing $\bar{C}_\chi = T^{-1}\mathbb{E}\sum_t \mathcal{I}_\chi$ and $\bar{L} = T^{-1}L$ makes the scaling visible:*

$$\textit{information side} = T\bar{C}_\chi, \qquad \textit{regret side} = Tr.$$

Thus T appears exactly as in one-round reductions, but no posterior trajectory has been collapsed into a static action distribution.

6.3 Bellman-Fano lower certificates

Theorem 6.1 is algorithm-specific. We now give algorithm-uniform certificates. The first bounds the information of any algorithm; the second bounds the ghost probability of success of any algorithm.

Exact information Bellman value and supersolutions. The intrinsic algorithm-uniform information quantity is an exact Bellman value. For a function F on states, define

$$(\mathsf{C}_t F)(s) := \sup_{p \in \Delta(\mathcal{A}_t(s))} \left\{ \mathcal{I}_\chi(s, p) + \mathbb{E}_{a\sim p, o \sim P_{s,a}} F(\tau(s, a, o)) \right\}.$$

Set

$$C^\star_{T+1}(s) = 0, \qquad C^\star_t(s) = (\mathsf{C}_t C^\star_{t+1})(s).$$

Then $C^\star_1(s_1)$ is the exact posterior-reference indexed-information capacity on the retained state. A sequence $\overline{C}_t$ is a valid information supersolution if

$$\overline{C}_{T+1} \ge 0, \qquad \overline{C}_t(s) \ge (\mathsf{C}_t \overline{C}_{t+1})(s) \quad \forall t, s.$$

For every algorithm starting from s_1,

$$C_\chi^{\mathsf{Alg}}(\mu) \le C^\star_1(s_1) \le \overline{C}_1(s_1). \tag{63}$$

Thus the exact Bellman recursion is the tight intrinsic object; supersolutions are used only as computable or analytically convenient upper certificates.

Exact ghost-good Bellman value and supersolutions. Let $m \in \Delta(\mathcal{Y})$ be the target-index distribution used to evaluate success, let s be the posterior-reference state, and let $g : \mathcal{Y} \to \mathbb{R}$ be an accumulated average-loss profile. For threshold r, define the exact ghost-good Bellman value by

$$\Gamma^{r,\star}_{T+1}(m, s, g) = m\{y : g(y) \le r\},$$

$$\Gamma^{r,\star}_{t}(m, s, g) = \sup_{p \in \Delta(\mathcal{A}_t(s))} \mathbb{E}_{a\sim p, o\sim P_{s,a}} \Gamma^{r,\star}_{t+1}\left(m, \tau(s, a, o), g + \frac{1}{T}\ell^{\chi}_{\cdot}(s, a)\right),$$

where $\ell^{\chi}_{\cdot}(s, a)$ is the function $y \mapsto \ell^{\chi}_{y}(s, a)$. A function sequence $\overline{\Gamma}^{r}_{t}$ is a valid ghost-mass supersolution if it dominates the terminal condition and the same Bellman right-hand side with $\overline{\Gamma}^{r}_{t+1}$ in place of $\Gamma^{r,\star}_{t+1}$. Then, for every algorithm,

$$p^{\mathsf{Alg}}_{r}(\mu, \chi) \le \Gamma^{r,\star}_{1}(\mu_\chi, s_1, 0) \le \overline{\Gamma}^{r}_{1}(\mu_\chi, s_1, 0). \tag{64}$$

The exact ghost entropy and a certified ghost entropy are therefore

$$\mathcal{E}^{r}_{\star,1}(\mu, \chi) := -\log \Gamma^{r,\star}_{1}(\mu_\chi, s_1, 0), \qquad \underline{\mathcal{E}}^{r}_{1}(\mu, \chi) := -\log \overline{\Gamma}^{r}_{1}(\mu_\chi, s_1, 0). \tag{65}$$

The certified entropy is conservative: $\underline{\mathcal{E}}^{r}_{1} \le \mathcal{E}^{r}_{\star,1}$.

Theorem 6.3 (Bellman quantile-index lower certificate)**.** *Assume the cumulative indexed loss is nonnegative as in Theorem 6.1. Suppose $\overline{C}_t$ is an information supersolution and $\overline{\Gamma}^{r}_{t}$ is a ghost-good-mass supersolution; the exact choices $\overline{C} = C^\star$ and $\overline{\Gamma} = \Gamma^{r,\star}$ give the tight intrinsic Bellman-Fano values. The statement is for the unrestricted nonanticipating class; for a Bellman-compatible restricted class, replace each local supremum in the exact recursions and in the supersolution recursions by the corresponding local feasible set. Then*

$$\inf_{\mathsf{Alg} \in \mathfrak{A}^{\mathrm{na}}_{T}} \mathbb{E}_{\Omega\sim\mu} \mathbb{E}_{\mathbb{P}^{\mathsf{Alg}}_{\Omega}} L_\Omega(H_T) \ge Tr\left[1 - \psi\big(e^{-\underline{\mathcal{E}}^{r}_{1}(\mu,\chi)}, \overline{C}_1(s_1)\big)\right]. \tag{66}$$

In particular, if

$$\overline{C}_1(s_1) + \log 2 \le \frac{1}{2}\underline{\mathcal{E}}^{r}_{1}(\mu, \chi), \tag{67}$$

then

$$\inf_{\mathsf{Alg} \in \mathfrak{A}^{\mathrm{na}}_{T}} \mathbb{E}_{\Omega\sim\mu} \mathbb{E}_{\mathbb{P}^{\mathsf{Alg}}_{\Omega}} L_\Omega(H_T) \ge \frac{Tr}{2}. \tag{68}$$

Consequently,

$$\mathfrak{R}^{\star}_{T} \ge \sup_{\mu,\chi,r}\left\{\frac{Tr}{2} : \overline{C}_1(s_1) + \log 2 \le \frac{1}{2}\underline{\mathcal{E}}^{r}_{1}(\mu, \chi)\right\}.$$

Proof. For every algorithm, (63) gives $T\bar{C}^{\mathsf{Alg}}_{\chi}(\mu) = C^{\mathsf{Alg}}_{\chi}(\mu) \le \overline{C}_1(s_1)$, and (64) gives $p^{\mathsf{Alg}}_{r}(\mu, \chi) \le e^{-\underline{\mathcal{E}}^{r}_{1}(\mu,\chi)}$. Substitute these two inequalities into Theorem 6.1, using the monotonicity of ψ in both arguments. The minimax statement follows from Yao's principle. □

The critical average-regret radius is

$$r_\star(\mu, \chi) := \sup\left\{r : \overline{C}_1(s_1) + \log 2 \le \frac{1}{2}\underline{\mathcal{E}}^{r}_{1}(\mu, \chi)\right\}. \tag{69}$$

The lower bound is order $Tr_\star$. A two-point Le Cam proof corresponds to the special case in which $\underline{\mathcal{E}} = O(1)$. A Fano or local-prior-mass lower bound has $\underline{\mathcal{E}}$ of order dimension or effective dimension.

6.4 DEC as one-step relaxation

As discussed in Section 5, offset DEC is obtained from the fixed-truth log-potential bracket (35) by making a one-step relaxation: the coordinate potential drop, or its posterior-averaged Bellman continuation counterpart, is replaced by a static one-step separation penalty, and the posterior-reference law is replaced by a fixed one-step reference predictive law.

In the model-index case $\chi(\omega) = \omega$, let $\mathfrak{C}_t(s)$ be the local comparison set at the current time-state pair and let $\bar{\nu}$ be a reference belief at state s. We write

$$P_{\bar{\nu},s,a} := \int P_{\omega,s,a}\, \bar{\nu}(d\omega)$$

for its predictive law. The corresponding pointwise KL offset is

$$\mathrm{dec}_{\gamma}^{\mathrm{KL}}(s;\bar{\nu}) := \inf_{p\in\Delta(\mathcal{A}_t(s))} \sup_{\omega\in\mathfrak{C}_t(s)} \left\{\mathbb{E}_{a\sim p}\ell_{\omega}(s,a) - \gamma\mathbb{E}_{a\sim p}D_{\mathrm{KL}}(P_{\omega,s,a} \,\|\, P_{\bar{\nu},s,a})\right\}. \tag{one-step model-index offset}$$

Thus the usual offset DEC is a worst-case, static-reference relaxation of the Bellman program: the learner chooses the one-step decision distribution p, and the local adversary then chooses the hardest model relative to the fixed reference predictive law.

The same one-step logic also appears on the lower-bound side, but only after an additional reduction from a dynamic posterior-reference experiment to a one-step reference experiment. Theorem 6.1 and Theorem 6.3 retain the dynamic reference posteriors. A one-step quantile DEC is obtained only by relaxing this dynamic proof: one fixes a reference model, or more generally a reference predictive law, decomposes the trajectory divergence into one-step divergences, and replaces the algorithm-dependent ghost history by a one-round decision distribution. In Chen et al. (2024, Section 3.2.2), this idea is made explicit through the quantile PAC-DEC. A reference model $\bar{M}$ generates the ghost experiment, an adversarial model M supplies the hard alternative, and the loss certificate is a quantile of the one-step risk rather than its expectation. The further reduction to constrained regret-DEC in that paper is precisely where localization, regularity, and problem-specific assumptions enter.

The passage from the dynamic theorem to a one-step coefficient is therefore not merely notational. On both the upper- and lower-bound sides, existing DEC measures can be viewed as one-step relaxations obtained by freezing the posterior-reference process into reference models or predictive laws. This interpretation is primarily conceptual: even the distinction between quantile and expected loss raises nontrivial technical questions when deriving regret-DEC lower bounds under regularity conditions. Nevertheless, the present paper adopts this viewpoint because it clarifies how one-step DEC certificates arise from the dynamic indexed Bellman picture.

Thus DEC should be read as a one-step relaxation of indexed Bellman information complexity, rather than as a universally tight conversion coefficient from information to regret. When the posterior geometry is simple, this relaxation can be sharp. In more structured problems, however, a one-round coefficient may obscure the dynamic object either by double-counting an entropy or estimation cost already accounted for by the posterior-reference lower bound, or by discarding the reference-trajectory geometry needed for localization. This issue appears in several examples: global RKHS DEC can be vacuous without localization, as discussed at the end of Section 7.1, or without a critical-radius regularity condition (Foster et al., 2023); in multi-armed bandits, an E2D upper bound may charge the cardinality complexity twice; in linear bandits, collapsing the adaptive posterior trajectory to a static action law can introduce an artificial gap in the lower bound; and for Bellman rank, DEC-style questions become intrinsic only after the relevant observable witnesses and their estimation complexity are fixed (Jiang et al., 2017; Du et al., 2021). The examples in Section 7 should therefore be viewed as indexed-Bellman certificates and diagnostics for when one-step DEC minimax calculation can be improved.

7 Applications and extensions

7.1 Finite-action kernel bandits and the four algorithms

This subsection specializes the indexed Bellman language to kernel bandits on a finite active action set. The point is not to make the unknown function finite-dimensional. The truth remains an element of an RKHS. What becomes finite is the marginal experiment observed through the active actions. This finite marginal is enough to define the GP posterior update, the model-index MAIR information, and the action-index AIR belief over the optimal active action. We present the exact finite-marginal indexed Bellman program first, and then compare GP–UCB, GP–E2D, and AMS/EBO as tractable relaxations or certificates of its bracket.

Problem and finite active marginal. Let $\mathcal{H}_k$ be a reproducing kernel Hilbert space (RKHS) on a possibly infinite domain $\mathcal{X}_{\text{all}}$, with kernel k satisfying $k(x,x) \le \kappa^2$. The learner is evaluated on a finite active set

$$\mathcal{X} = \{x_1, \ldots, x_n\} \subset \mathcal{X}_{\text{all}}.$$

The unknown reward function $f^\star \in \mathcal{H}_k$ satisfies $\|f^\star\|_{\mathcal{H}_k} \le B$, and at round t the learner chooses $X_t \in \mathcal{X}$ and observes

$$Y_t = f^\star(X_t) + \varepsilon_t, \qquad \mathbb{E}[\varepsilon_t \mid H_{t-1}, X_t] = 0, \tag{70}$$

where the noise is conditionally R-sub-Gaussian. For the Gaussian calculations below we use the algorithmic reference model with variance parameter $\lambda > 0$; the frequentist confidence event is obtained by the usual self-normalized argument.

Let $F = (f(x_1), \ldots, f(x_n)) \in \mathbb{R}^n$ be the active reward vector and let $K_{\mathcal{X}}$ be the $n \times n$ kernel matrix. The GP reference prior is

$$F \sim N(0, K_{\mathcal{X}}), \qquad Y_t \mid F, X_t = x_i \sim N(F_i, \lambda).$$

After history H_{t-1}, the algorithmic posterior on F is Gaussian,

$$F \mid H_{t-1} \sim N(m_t, \Sigma_t).$$

For $x_i \in \mathcal{X}$, write $m_t(x_i) = e_i^\top m_t$ and $s_t^2(x_i) = e_i^\top \Sigma_t e_i$. If $X_t = x_i$, the exact posterior update is

$$m_{t+1} = m_t + \frac{\Sigma_t e_i}{\lambda + s_t^2(x_i)} \{Y_t - m_t(x_i)\}, \tag{71}$$

$$\Sigma_{t+1} = \Sigma_t - \frac{\Sigma_t e_i e_i^\top \Sigma_t}{\lambda + s_t^2(x_i)}. \tag{72}$$

This is a finite marginal posterior update. It should not be read as an assumption that $f^\star$ has n parameters. The RKHS function may be infinite-dimensional; only the observations and decisions in this finite experiment depend on the vector F.

Model-index information and action-index information. For the model-index specialization, take the index to be the active reward vector $Y_{\mathrm{MAIR}} = F$. The one-step GP/MAIR information at state H_{t-1} is

$$\mathcal{I}_t^{\mathrm{GP}}(p) := \mathbb{E}_{x\sim p}\frac{1}{2}\log\left(1+\frac{s_t^2(x)}{\lambda}\right). \tag{73}$$

Along a realized action sequence,

$$C_T^{\mathrm{GP}} := \sum_{t=1}^{T}\mathcal{I}_t^{\mathrm{GP}}(\delta_{X_t}) = \frac{1}{2}\log\det\bigl(I+\lambda^{-1}K_{X_{1:T},X_{1:T}}\bigr), \tag{74}$$

where repeated actions are included in the Gram matrix. This is the realized information gain of the algorithmic trajectory. It can be much smaller than the maximal information gain that maximizes over all length-T designs.

For the action-index specialization, assume ties are broken by a fixed rule and set

$$A^\star(F) \in \arg\max_{x\in\mathcal{X}} F_x, \qquad q_t(a) := \mathbb{P}(A^\star = a \mid H_{t-1}).$$

Let ν_t^a be the exact conditional law of F given H_{t-1} and $A^\star = a$. It is generally a truncated Gaussian on the cone where a is optimal. Define the conditional predictive law

$$P_{t,a,x}(\cdot) := \int N(F_x,\lambda)(\cdot)\,\nu_t^a(dF),$$

for the observation at action x given the event $A^\star = a$. The marginal predictive is

$$P_{t,x} = \sum_{a\in\mathcal{X}} q_t(a)P_{t,a,x} = N(m_t(x), \lambda + s_t^2(x)).$$

The AIR information increment is

$$\mathcal{I}_t^{\mathrm{AIR}}(p) := \mathbb{E}_{x\sim p}\sum_{a\in\mathcal{X}} q_t(a)D_{\mathrm{KL}}(P_{t,a,x}\|P_{t,x}). \tag{75}$$

The conditional AIR regret is

$$\Delta_t^{\mathrm{AIR}}(p) := \sum_{a\in\mathcal{X}} q_t(a)\mathbb{E}_{F\sim\nu_t^a}\mathbb{E}_{x\sim p}[F_a - F_x]. \tag{76}$$

The exact indexed chain rule gives

$$\sum_{t=1}^{T}\mathbb{E}\mathcal{I}_t^{\mathrm{AIR}}(p_t) = I(A^\star; H_T) \le H(A^\star) \le \log n.$$

Thus AIR places the entropy telescope on the decision index rather than on the ambient RKHS. The price is that the coefficient converting action-index information into regret must be controlled by an algorithmic bracket.

Algorithm 1: finite-marginal AIR/MAIR Bellman DP. On the finite active marginal, the exact frequentist benchmark is the log-penalized Bellman program of Theorem 5.2. The state contains (m_t, Σ_t, q_t), where (m_t, Σ_t) is the algorithmic GP marginal on F and q_t is the maintained reference law of the index, either $A^\star(F)$ for AIR or F for MAIR. Let $\mathfrak{C}_t(m, \Sigma, q)$ be a calibrated local set of candidate active reward vectors containing the fixed truth $F^\star$ on the calibration event. In the action-index version define

$$y(f) = A^\star(f), \qquad \mathsf{G}_t^{\mathrm{AIR}}(p; f) := \mathbb{E}_{x\sim p,\, O\sim N(f_x,\lambda)} \log \frac{q^+(y(f) \mid m, \Sigma, q, x, O)}{q(y(f))},$$

where q^+ is the chosen reference update for the optimal-action marginal. The robust finite-marginal Bellman recursion is

$$W_t^\gamma(m, \Sigma, q) = \inf_{p\in\Delta(\mathcal{X})} \sup_{f\in\mathfrak{C}_t(m,\Sigma,q)} \left\{\Delta_f(p) + \mathbb{E}_f W_{t+1}^\gamma(m^+, \Sigma^+, q^+) - \gamma \mathsf{G}_t^{\mathrm{AIR}}(p; f)\right\}.$$

The model-index MAIR version replaces $y(f)$ by the finite reward vector f and uses the corresponding model-coordinate posterior log gain. By Theorem 5.2, an approximate minimizer of this recursion has regret at most

$$W_1^\gamma(m_1, \Sigma_1, q_1) + \gamma \log \frac{1}{q_1(A^\star(F^\star))} + \text{optimization and calibration errors.}$$

If the maintained belief is the exact Bayesian posterior and one averages this coordinate recursion over F, the robust supremum is replaced by posterior expectation and the dynamic program reduces to the posterior-averaged AIR recursion

$$U_t^\gamma(m, \Sigma, q) = \inf_{p\in\Delta(\mathcal{X})} \left\{\Delta_t^{\mathrm{AIR}}(p) + \mathbb{E} U_{t+1}^\gamma(m^+, \Sigma^+, q^+) - \gamma \mathcal{I}_t^{\mathrm{AIR}}(p)\right\}.$$

Thus the finite-marginal Bellman DP is the clean upper object for comparison with the lower theorem: the frequentist version uses the coordinate log potential and calibrated candidate truths, while the Bayesian version is its posterior average. The following three rules are tractable certificates or relaxations of this benchmark.

Algorithm 2: GP–UCB. GP–UCB uses the Gaussian marginal posterior (m_t, Σ_t) on the finite active marginal. In this paragraph, e_x denotes the coordinate vector associated with $x \in \mathcal{X}$, and $s_t(x) = (e_x^\top \Sigma_t e_x)^{1/2}$ denotes the unmultiplied posterior standard deviation, i.e., the generic uncertainty scale $\sigma_t(x)$ from Lemma 5.4. The confidence half-width is

$$w_t(x) := \beta_t s_t(x),$$

where β_t is the confidence multiplier. The algorithm chooses

$$X_t \in \arg\max_{x\in\mathcal{X}} \{m_t(x) + \beta_t s_t(x)\}. \tag{77}$$

The following event is the frequentist calibration certificate:

$$\mathcal{E}_{\mathrm{GP}} := \{|f^\star(x) - m_t(x)| \le w_t(x) = \beta_t s_t(x) \quad \text{for all } t \le T, x \in \mathcal{X}\}. \tag{78}$$

For finite $\mathcal{X}$, standard RKHS self-normalized concentration gives $\mathbb{P}(\mathcal{E}_{\mathrm{GP}}) \ge 1 - \delta$ for the usual choice of β_t. For example, up to universal constants one may take

$$\beta_t \asymp B + R\sqrt{\Gamma_{t-1}^{\mathrm{GP}} + \log(1/\delta)}, \qquad \Gamma_m^{\mathrm{GP}} := \sup_{x_{1:m}\in\mathcal{X}^m} \frac{1}{2} \log\det\left(I + \lambda^{-1} K_{x_{1:m},x_{1:m}}\right),$$

with the precise form depending on the normalization of k and the ridge parameter. Here $K_{x_{1:m},x_{1:m}}$ is the Gram matrix of the possibly repeated design sequence $(x_1, \dots, x_m)$. The quantity Γ_m^{GP} is a worst-case calibration device for the confidence event; it is different from the realized information gain C_T^{GP} paid in the regret bound. The next theorem isolates the deterministic part of the GP–UCB proof.

Theorem 7.1 (GP–UCB realized-information bound)**.** *On the event $\mathcal{E}_{\mathrm{GP}}$, the GP–UCB rule (77) satisfies*

$$\mathrm{Reg}_T(f^\star) \le 2\beta_T \sum_{t=1}^{T} s_t(X_t) \le 2\beta_T \sqrt{T\, c_{\kappa,\lambda}\, C_T^{\mathrm{GP}}}, \tag{79}$$

where

$$c_{\kappa,\lambda} := \frac{2\kappa^2}{\log(1+\kappa^2/\lambda)}$$

and C_T^{GP} is the realized information gain in (74). Consequently, if rewards are bounded in $[0,1]$ and $\mathbb{P}(\mathcal{E}_{\mathrm{GP}}) \ge 1-\delta$, then

$$\mathbb{E}\,\mathrm{Reg}_T(f^\star) \le 2\beta_T \sqrt{T\, c_{\kappa,\lambda}\, \mathbb{E} C_T^{\mathrm{GP}}} + T\delta.$$

Proof. Let $x^\star \in \arg\max_{x \in \mathcal{X}} f^\star(x)$. On $\mathcal{E}_{\mathrm{GP}}$, calibration at $x^\star$ and the optimistic choice of X_t give

$$f^\star(x^\star) \le m_t(x^\star) + w_t(x^\star) \le m_t(X_t) + w_t(X_t).$$

Calibration at X_t gives $m_t(X_t) \le f^\star(X_t) + w_t(X_t)$. Hence

$$f^\star(x^\star) - f^\star(X_t) \le 2w_t(X_t) = 2\beta_t s_t(X_t) \le 2\beta_T s_t(X_t).$$

Summing and applying Cauchy–Schwarz yields

$$\mathrm{Reg}_T(f^\star) \le 2\beta_T \sqrt{T \sum_{t=1}^{T} s_t^2(X_t)}.$$

For $u \in [0, \kappa^2/\lambda]$, monotonicity of $u/\log(1+u)$ on a bounded interval gives

$$\lambda u \le \frac{\kappa^2}{\log(1+\kappa^2/\lambda)} \log(1+u).$$

Applying this with $u = s_t^2(X_t)/\lambda$ and using (74) proves (79). The expectation bound adds the trivial regret bound T on $\mathcal{E}_{\mathrm{GP}}^c$ and uses Jensen's inequality. □

GP–UCB as a MAIR decomposition. The theorem is Lemma 5.4 with the generic uncertainty scale $\sigma_t(X_t) = s_t(X_t)$ and the realized-information log determinant telescope

$$\sum_t \frac{1}{2} \log\left(1 + s_t^2(X_t)/\lambda\right) = C_T^{\mathrm{GP}}.$$

The confidence half-width is $w_t(X_t) = \beta_t s_t(X_t)$, but the multiplier β_t is not part of the information telescope; it is the coefficient paid for calibration and optimism in the fixed-truth bracket. This separation is the reason the proof uses both s_t and w_t: s_t^2 is summed by the log-determinant/elliptical-potential certificate, whereas w_t is used only to upper bound instantaneous regret. GP–UCB is often practically strong because it follows the realized trajectory and pays the realized information gain C_T^{GP}, not the worst possible design information Γ_T^{GP}, even though its confidence multiplier may be conservative in step-uniform minimax analyses.

Algorithm 3: GP–E2D. A GP–E2D rule keeps the same GP posterior reference but chooses a distribution over actions by solving a localized DEC offset. Given a localized comparison set $\mathcal{F}_t$ containing plausible functions, define

$$\Delta_f(p) := \max_{x \in \mathcal{X}} f(x) - \mathbb{E}_{x \sim p} f(x),$$

and let $P_{f,x} = N(f(x), \lambda)$ while $P_{t,x} = N(m_t(x), \lambda + s_t^2(x))$ is the posterior predictive. A schematic GP–E2D decision is

$$p_t \in \arg\min_{p \in \Delta(\mathcal{X})} \sup_{f \in \mathcal{F}_t} \left\{ \Delta_f(p) - \gamma_t \mathbb{E}_{x \sim p} D_{\mathrm{KL}}(P_{f,x} \| P_{t,x}) \right\}. \tag{80}$$

If $f^\star \in \mathcal{F}_t$ and the optimized value in (80) is at most ε_t, then

$$\Delta_{f^\star}(p_t) \le \gamma_t \mathbb{E}_{x \sim p_t} D_{\mathrm{KL}}(P_{f^\star,x} \| P_{t,x}) + \varepsilon_t. \tag{81}$$

Summing (81) gives a regret bound once the fixed-truth KL terms are related to a posterior-ratio or information telescope. Under a well-specified Bayesian GP reference this relation holds in expectation by the MAIR chain rule. Under a fixed frequentist truth it requires calibration/localization. This is the technical point that a GP–E2D analysis must address: the one-round DEC is optimized, but the compatibility between that coefficient and the realized GP log determinant is not automatic. An unlocalized offset over an infinite-dimensional RKHS ball can be unbounded or vacuous; meaningful GP–E2D guarantees must localize the comparison set to the posterior-reference experiment.

Algorithm 4: AMS/EBO on the action index. AMS/EBO attacks the same specialized log-potential bracket from the robust posterior side. In the finite-action kernel marginal, the natural AIR index is $A^\star$. Let q_t be the current reference distribution on $A^\star$, let $\mathfrak{B}_t(q_t)$ be a calibrated local set of candidate beliefs ν on the active reward vector F, and fix a constant coefficient $\gamma > 0$. For each ν, define $\Delta_\nu^{\mathrm{AIR}}(p)$ and $\mathcal{I}_\nu^{\mathrm{AIR}}(p)$ by the analogues of (76) and (75), using the conditional laws induced by ν. A robust AIR/EBO decision is

$$p_t \in \arg\min_{p \in \Delta(\mathcal{X})} \sup_{\nu \in \mathfrak{B}_t(q_t)} \left\{ \Delta_\nu^{\mathrm{AIR}}(p) - \gamma \mathcal{I}_\nu^{\mathrm{AIR}}(p) - \gamma D_{\mathrm{KL}}(\nu_{A^\star} \| q_t) \right\}. \tag{82}$$

The maximization over ν is essential: without it the display is only the posterior AIR bracket for a chosen belief, not the robust AMS/EBO relaxation. In a finite-index abstraction, the last two terms are the KL-dual form of the log-partition potential (40). If the robust value in (82) is at most ε_t, the required concavity/first-order conditions in Lemma 5.5 hold, and the belief set is calibrated so that the fixed truth $f^\star$ is an admissible comparison direction up to error cal_T, then Theorem 5.1 gives

$$\mathbb{E}_{f^\star} \mathrm{Reg}_T \le \gamma \log \frac{1}{q_1(a^\star)} + \mathbb{E}_{f^\star} \sum_{t=1}^{T} \varepsilon_t + \mathrm{cal}_T, \tag{83}$$

where $a^\star = A^\star(f^\star)$. The constant coefficient avoids the false shortcut of replacing a weighted fixed-truth log-gain sum by a single telescope. If a varying coefficient is used, the corresponding weighted telescope must be proved separately. Under a Bayesian reference prior, posterior averaging of the same coordinate bound gives

$$\mathbb{E}\, \mathrm{Reg}_T \le \gamma H(A^\star) + \mathbb{E} \sum_{t=1}^{T} \varepsilon_t + \mathbb{E}\, \mathrm{cal}_T\,.$$

A ratio form gives the familiar Cauchy–Schwarz variant after replacing the fixed coefficient by the appropriate sequential AIR ratio and applying the action-index chain rule. Since $H(A^\star) \le \log n$, this route scales with the finite decision index and the sequential AIR coefficient. It is different from GP–E2D: AMS/EBO uses a robust action-index belief trajectory, whereas GP–E2D optimizes a one-round model-index offset whose relationship to the realized log determinant or posterior-ratio telescope must be justified separately.

Position relative to the GP–UCB optimality debate. The original GP–UCB analysis is practical and trajectory-sensitive because it pays the realized information gain (74), or a worst-case relaxation only at the final step (Srinivas et al., 2010). Later work asks whether the remaining suboptimality comes from the algorithm or from the confidence analysis; in particular, Vakili et al. (2021b) ask for sharper online RKHS confidence intervals and sharper finite-action kernel-bandit guarantees. The indexed view separates the issue into four mechanisms. The finite-marginal indexed Bellman DP is the exact information-potential benchmark, but may be computationally demanding. GP–UCB has the strongest realized-information geometry but may have a conservative MAIR coefficient. AMS/EBO can keep the action-index AIR telescope $I(A^\star; H_T) \le \log n$ and therefore has a principled route to finite-decision complexity, provided the robust bracket can be controlled. GP–E2D directly optimizes a DEC coefficient, but an unlocalized DEC can be vacuous and a localized DEC must be matched to the same posterior-reference trajectory. Thus these algorithms should not be compared by a single universal coefficient; they are one exact Bellman benchmark and three relaxations or certificates of the same indexed Bellman identity, with different estimation and coefficient mechanisms.

Why global RKHS DEC can be vacuous without suitable localization. Consider a Gaussian RKHS bandit with noise variance σ^2, feature map $\phi(x) \in \mathcal{H}$ satisfying $\|\phi(x)\|_{\mathcal{H}} \le 1$, and suppose the action set contains points $\{x_j : j \ge 1\}$ with $\phi(x_j) = e_j$, where $\{e_j\}_{j\ge 1}$ is orthonormal in $\mathcal{H}$. Let the reference model be $f_0 \equiv 0$, and let $\mathcal{F}_B = \{f \in \mathcal{H} : \|f\|_{\mathcal{H}} \le B\}$. For any one-round decision distribution p, define

$$q_j := \mathbb{E}_{x\sim p}\langle e_j, \phi(x)\rangle_{\mathcal{H}}^2.$$

Since

$$\sum_{j\ge 1} q_j \le \mathbb{E}_{x\sim p}\|\phi(x)\|_{\mathcal{H}}^2 \le 1,$$

there are directions with arbitrarily small q_j. For the alternative $f_j = Be_j$, the optimal action is x_j, while

$$\mathbb{E}_{x\sim p}\Delta_{f_j}(x) = B - B\,\mathbb{E}_{x\sim p}\langle e_j, \phi(x)\rangle_{\mathcal{H}} \ge B(1 - \sqrt{q_j}),$$

and the one-round KL information against f_0 is

$$\mathbb{E}_{x\sim p} D_{\mathrm{KL}}\big(N(f_j(x), \sigma^2) \,\big\|\, N(0, \sigma^2)\big) = \frac{B^2}{2\sigma^2} q_j.$$

Hence, for every finite offset coefficient γ, one may choose j with q_j so small that

$$\mathbb{E}_{x\sim p}\Delta_{f_j}(x) - \gamma\mathbb{E}_{x\sim p} D_{\mathrm{KL}}\big(N(f_j(x), \sigma^2) \,\big\|\, N(0, \sigma^2)\big) \ge cB$$

for a numerical constant $c > 0$. The constrained version has the same pathology: for any fixed information radius, a sufficiently rare direction has information inside the radius while retaining regret of order B. Thus an unlocalized DEC over the entire RKHS ball need not reflect the effective difficulty of the finite experiment being run. The point is that the localization is part of the representation and posterior-reference trajectory; it should not be appended as an independent estimation term after a separate log-determinant analysis.

7.2 MAB lower bound via optimal-action index and reference history

In this subsection, we apply our lower-bound approach to multi-armed bandits (MAB). Consider $K \geq 3$ Gaussian arms with unit noise variance. Let the ambient model space be a class $\Omega_{\text{MAB}} \subseteq \mathbb{R}^K$ of mean vectors, and write a model environment as $\omega = (\theta_\omega(1), \ldots, \theta_\omega(K))$. Pulling arm a in environment ω produces an observation distributed as $N(\theta_\omega(a), 1)$. Let

$$A^\star(\omega) \in \arg\max_{a \in [K]} \theta_\omega(a)$$

with an arbitrary fixed tie-breaking rule, and define the information index

$$\chi(\omega) = A^\star(\omega) \in [K].$$

Thus the random environment is the full mean vector Ω, while the information target used in the lower certificate is only

$$Y = \chi(\Omega) = A^\star(\Omega).$$

For the lower bound, fix $\Delta > 0$ and use the K-point prior μ_Δ supported on the spike environments

$$\omega^j = \Delta e_j, \qquad j \in [K],$$

where e_j is the jth coordinate vector. Equivalently, draw $J \sim \mathrm{Unif}([K])$ and set $\Omega = \omega^J$. On this prior support the optimal arm is unique and

$$Y = \chi(\Omega) = J.$$

The equality $Y = J$ is only a property of this least-favorable prior; it should not be read as identifying the information target with the full model environment. The one-step regret in environment ω^j is

$$\ell_{\omega^j}(a) = \max_b \theta_{\omega^j}(b) - \theta_{\omega^j}(a) = \Delta \mathbb{1}\{a \neq j\}.$$

For brevity write $\ell_j = \ell_{\omega^j}$ on this subfamily.

Lemma 7.2 (MAB ghost entropy)**.** *Let $Y = \chi(\Omega)$ under the prior $\Omega \sim \mu_\Delta$. For every algorithm and every reference history H'_T independent of $Y \sim \mathrm{Unif}([K])$,*

$$\mathbb{P}\left(\bar{L}_Y(H'_T) \leq \frac{\Delta}{2}\right) \leq \frac{2}{K},$$

where $\bar{L}_y(H'_T) = T^{-1} \sum_{t=1}^T \ell_y(A'_t)$ and A'_t denotes the arm selected in the reference history at time t.

Proof. For a fixed reference action sequence $a'_1, \ldots, a'_T$, let

$$N'_y = \sum_{t=1}^T \mathbb{1}\{a'_t = y\}.$$

The average regret against the spike environment ω^y is

$$\bar{L}_y(H'_T) = \Delta(1 - N'_y/T).$$

The event $\bar{L}_y(H'_T) \leq \Delta/2$ implies $N'_y \geq T/2$. Since Y is uniform and independent of the reference sequence,

$$\mathbb{E}[N'_Y \mid H'_T] = \frac{1}{K}\sum_{y=1}^{K} N'_y = \frac{T}{K}.$$

Markov's inequality gives

$$\mathbb{P}(N'_Y \geq T/2 \mid H'_T) \leq \frac{2}{K}.$$

Averaging over H'_T proves the claim. □

Lemma 7.3 (MAB information bound)**.** *There is a universal constant $c_0 > 0$ such that if $\Delta^2 T/K \leq c_0$, then every adaptive algorithm satisfies*

$$I(Y; H_T) \leq \frac{1}{4}\log(K/2),$$

where $Y = \chi(\Omega)$ and $\Omega \sim \mu_\Delta$.

Proof. Let $\mathbb{P}_0$ be the law of the history under the zero-mean reference model, let $\mathbb{P}_j$ be the law under the spike environment ω^j, and let

$$\bar{\mathbb{P}} = \frac{1}{K}\sum_{j=1}^{K} \mathbb{P}_j$$

be the marginal law of H_T under the prior μ_Δ. Under this prior, $Y = j$ is equivalent to $\Omega = \omega^j$, but the mutual information is still the action-index information $I(Y; H_T)$, not information about an ambient full mean vector outside the prior support. Since $\bar{\mathbb{P}} \geq K^{-1}\mathbb{P}_j$, the likelihood ratio $d\mathbb{P}_j/d\bar{\mathbb{P}}$ is bounded by K. The inequality

$$D_{\mathrm{KL}}(P\|Q) \leq (2 + \log K)H^2(P, Q) \qquad \text{whenever } Q \geq K^{-1}P,$$

together with the triangle inequality and convexity for squared Hellinger distance, gives

$$I(Y; H_T) = \frac{1}{K}\sum_{j=1}^{K} D_{\mathrm{KL}}(\mathbb{P}_j\|\bar{\mathbb{P}}) \leq 4(2 + \log K)\frac{1}{K}\sum_{j=1}^{K} H^2(\mathbb{P}_j, \mathbb{P}_0).$$

Using $H^2(P, Q) \leq D_{\mathrm{KL}}(Q\|P)$ and the adaptive Gaussian KL chain rule under the zero model,

$$\frac{1}{K}\sum_{j=1}^{K} H^2(\mathbb{P}_j, \mathbb{P}_0) \leq \frac{1}{K}\sum_{j=1}^{K} D_{\mathrm{KL}}(\mathbb{P}_0\|\mathbb{P}_j) = \frac{\Delta^2}{2K}\mathbb{E}_0\sum_{t=1}^{T}\sum_{j=1}^{K} \mathbb{1}\{A_t = j\} = \frac{\Delta^2 T}{2K}.$$

Therefore

$$I(Y; H_T) \leq 2(2 + \log K)\frac{\Delta^2 T}{K}.$$

Choosing, for example,

$$c_0 \leq \inf_{K\geq 3} \frac{\log(K/2)}{8(2 + \log K)}$$

which is a strictly positive universal constant, proves the displayed bound. □

Theorem 7.4 (MAB lower bound)**.** *For Gaussian K-armed bandits with unit noise variance, over any ambient mean-vector class Ω_{MAB} that contains the spike instances $\{\Delta e_j : j \in [K]\}$ used below,*

$$\mathfrak{R}_T^\star \geq c\sqrt{KT}$$

for a universal constant $c > 0$ whenever $T \geq K \geq 3$. In particular, this applies to any bounded class such as $[0,1]^K$ once the universal constant in the choice of Δ is small enough.

Proof. Choose
$$\Delta = c_1\sqrt{K/T}$$
with c_1 small enough that Lemma 7.3 applies. Let $r = \Delta/2$. Lemma 7.2 gives $p_r \leq 2/K$, while Lemma 7.3 gives
$$T\bar{C} = I(Y; H_T) \leq \frac{1}{4}\log(K/2).$$
For c_1 small enough, condition (62) holds. Theorem 6.1, applied uniformly over algorithms, gives a Bayes regret lower bound under the prior μ_Δ supported on the spike subfamily. Since this prior is supported inside the ambient model class Ω_{MAB}, the minimax regret over Ω_{MAB} is at least this Bayes risk. Hence
$$\mathfrak{R}_T^\star \geq \frac{Tr}{2} = \frac{T\Delta}{4} = c\sqrt{KT}$$
for a universal constant $c > 0$. □

On the upper-bound side, the same distinction between the full model environment and the action index is important. An action-index AIR analysis uses $Y = \chi(\Omega) = A^\star(\Omega)$ as the information target, rather than the full environment Ω. This avoids charging information for distinctions between environments that do not change the optimal action. A model-index E2D relaxation can pay a K-factor both through model estimation and through the DEC term, whereas the action-index AIR identity in Theorem 4.6 incurs only a $\log K$ information-gain term and pays the K-factor through the AIR coefficient (Xu and Zeevi, 2025). This nearly matches the lower bound.

7.3 Linear bandits lower bound via optimal-action index and reference history

Consider stochastic linear bandits with action set

$$\mathcal{A} = \{a \in \mathbb{R}^d : \|a\|_2 \leq 1\}.$$

Let the ambient model class be the Euclidean unit parameter ball

$$\Omega_{\mathrm{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \leq 1\},$$

and regard an environment $\omega \in \Omega_{\mathrm{lin}}$ as the full mean parameter vector. At time t, after choosing $A_t \in \mathcal{A}$, the learner observes

$$O_t = \langle \omega, A_t \rangle + \xi_t, \qquad \xi_t \sim N(0,1),$$

where the noises are independent over time and independent of the learner's external randomization. For $\omega \neq 0$, the unique optimal action over $\mathcal{A}$ is

$$A^\star(\omega) = \frac{\omega}{\|\omega\|_2}.$$

At $\omega = 0$, fix any deterministic tie-breaking action; this case is not used by the lower-bound prior. The information index is the optimal action

$$\chi(\omega) = A^\star(\omega), \qquad Y = \chi(\Omega),$$

not the full environment Ω. Thus two environments that have the same optimal action are not distinguished by the index unless the prior support itself makes them distinguishable.

We prove the standard $\Omega(d\sqrt{T})$ minimax lower bound through the quantile indexed information theorem. The proof uses a hypercube prior together with an adaptive information bound. The role of the hypercube variable below is only to code the finitely many optimal actions in the prior support. It should not be read as replacing the ambient environment Ω by a model label. Note that for a non-Gaussian prior, one cannot condition on the realized adaptive design matrix and then apply the fixed-design Gaussian-channel capacity formula, because the design itself is a statistic of the past observations and may carry information about the unknown environment and hence about Y. Chen et al. (2024, Section 3.3) prove a closely related lower bound by a different route: they use the interactive Fano method with a Gaussian prior and a unit-sphere calculation. Both approaches show that action-indexed information arguments can recover the sharp linear-bandit order, whereas model-indexed relaxations such as DEC can lose factors when they charge for estimating more of the model than is needed to identify the optimal action.

Let

$$V \sim \mathrm{Unif}(\{\pm 1\}^d), \qquad \Omega = \Delta V.$$

Assume $\Delta\sqrt{d} \le 1$, so that this prior is supported on Ω_{lin}. On this prior support,

$$Y = \chi(\Omega) = A^\star(\Omega) = \frac{V}{\sqrt{d}}. \tag{84}$$

The map $v \mapsto v/\sqrt{d}$ is one-to-one on $\{\pm 1\}^d$. Hence, for this prior only, V may be used as a code for the action index Y, and

$$I(Y; H_T) = I(V; H_T).$$

For an index value $y = v/\sqrt{d}$, write ℓ_v for the regret under the corresponding environment $\omega = \Delta v$. The one-step regret of action $a \in \mathbb{R}^d$, $\|a\|_2 \le 1$, is

$$\ell_v(a) = \Delta\sqrt{d} - \Delta\langle v, a\rangle.$$

Assume $T \ge d^2$ and choose $\Delta \le 1/\sqrt{d}$, so that $\|\Omega\|_2 = \Delta\sqrt{d} \le 1$.

Lemma 7.5 (Linear ghost entropy)**.** *There is a universal constant $c_{\mathrm{gh}} > 0$ such that every deterministic reference action sequence $a'_1, \ldots, a'_T$, with $\|a'_t\|_2 \le 1$, satisfies*

$$\mathbb{P}_{V\sim\mathrm{Unif}(\{\pm1\}^d)}\left(\frac{1}{T}\sum_{t=1}^{T}\big(\Delta\sqrt{d} - \Delta\langle V, a'_t\rangle\big) \le \frac{\Delta\sqrt{d}}{4}\right) \le \exp(-c_{\mathrm{gh}} d).$$

Consequently, the same bound holds after averaging over any random reference history H'_T whose induced action sequence is independent of the action index Y, equivalently independent of V under the hypercube prior.

Proof. Let

$$\bar{a} := \frac{1}{T}\sum_{t=1}^{T} a'_t.$$

Since $\|\bar{a}\|_2 \le T^{-1} \sum_t \|a'_t\|_2 \le 1$, the event in the display implies

$$\Delta\sqrt{d} - \Delta\langle V, \bar{a}\rangle \le \frac{\Delta\sqrt{d}}{4},$$

or equivalently,

$$\langle V, \bar{a}\rangle \ge \frac{3\sqrt{d}}{4}.$$

The random variable $\langle V, \bar{a}\rangle = \sum_{i=1}^d V_i \bar{a}_i$ is a centered Rademacher sum with variance proxy $\|\bar{a}\|_2^2 \le 1$. Hoeffding's inequality therefore gives

$$\mathbb{P}\left(\langle V, \bar{a}\rangle \ge \frac{3\sqrt{d}}{4}\right) \le \exp\left(-\frac{9d}{32}\right).$$

Thus the first claim holds, for example with $c_{\mathrm{gh}} = 9/32$. If the reference action sequence is random but independent of Y, then because $Y = V/\sqrt{d}$ is a bijective function of V on the prior support, the reference sequence is independent of V. Conditioning on the reference history and applying the deterministic bound gives the second claim. □

Lemma 7.6 (Adaptive hypercube information capacity)**.** *For the hypercube prior above, every possibly randomized adaptive algorithm satisfies*

$$I(Y; H_T) = I(V; H_T) \le \Delta^2 T,$$

where $H_T = (A_1, O_1, \dots, A_T, O_T)$.

Proof. Since $Y = V/\sqrt{d}$ is a bijective function of V on the hypercube support, $I(Y; H_T) = I(V; H_T)$. It remains to bound $I(V; H_T)$. Let U denote the learner's internal random seed, independent of V. Then

$$I(V; H_T) \le I(V; H_T, U) = \mathbb{E}_U I(V; H_T \mid U).$$

It is therefore enough to prove the claim after conditioning on U, so that the algorithm is deterministic.

For $v \in \{\pm 1\}^d$, let $\mathbb{P}_v$ denote the trajectory law when $\Omega = \Delta v$. We use the chain rule over coordinates:

$$I(V; H_T) = \sum_{i=1}^d I(V_i; H_T \mid V_1, \dots, V_{i-1}).$$

Fix a coordinate i and a prefix $u = (v_1, \dots, v_{i-1})$. Let $\mathbb{P}_{u,+}$ and $\mathbb{P}_{u,-}$ be the trajectory laws after averaging uniformly over the remaining coordinates $V_{i+1}, \dots, V_d$, conditional respectively on $V_i = +1$ and $V_i = -1$. Then

$$I(V_i; H_T \mid V_{<i} = u) = D_{\mathrm{JS}}(\mathbb{P}_{u,+}, \mathbb{P}_{u,-}),$$

where D_{JS} denotes Jensen–Shannon divergence with equal weights. For any two probability measures P, Q,

$$D_{\mathrm{JS}}(P, Q) \le \frac{1}{4}\Big(D_{\mathrm{KL}}(P\|Q) + D_{\mathrm{KL}}(Q\|P)\Big).$$

Next couple the two mixtures by using the same suffix. If $w \in \{\pm 1\}^{d-i}$ and

$$v^+(u, w) = (u, +1, w), \qquad v^-(u, w) = (u, -1, w),$$

then joint convexity of relative entropy gives

$$D_{\mathrm{KL}}(\mathbb{P}_{u,+}\|\mathbb{P}_{u,-}) \le \mathbb{E}_w D_{\mathrm{KL}}\big(\mathbb{P}_{v^+(u,w)} \,\big\|\, \mathbb{P}_{v^-(u,w)}\big)\,,$$

and the same argument gives the analogous reverse inequality.

For two fixed vertices v^+ and v^- that differ only in coordinate i, the adaptive KL chain rule gives

$$\begin{aligned}
D_{\mathrm{KL}}(\mathbb{P}_{v^+}\|\mathbb{P}_{v^-}) &= \frac{1}{2}\mathbb{E}_{v^+}\sum_{t=1}^{T}\Big(\Delta\langle v^+ - v^-, A_t\rangle\Big)^2 \\
&= 2\Delta^2\mathbb{E}_{v^+}\sum_{t=1}^{T} A_{t,i}^2.
\end{aligned}$$

Here the action kernel contributes no KL because, after conditioning on the learner's internal randomness, the algorithm uses the same decision rule under both environments; only the Gaussian observation kernel changes. Similarly,

$$D_{\mathrm{KL}}(\mathbb{P}_{v^-}\|\mathbb{P}_{v^+}) = 2\Delta^2\mathbb{E}_{v^-}\sum_{t=1}^{T} A_{t,i}^2.$$

Combining the preceding displays yields

$$\begin{aligned}
I(V_i; H_T \mid V_{<i} = u) &\le \frac{\Delta^2}{2}\mathbb{E}_w\left[\mathbb{E}_{v^+(u,w)}\sum_{t=1}^{T} A_{t,i}^2 + \mathbb{E}_{v^-(u,w)}\sum_{t=1}^{T} A_{t,i}^2\right] \\
&= \Delta^2\mathbb{E}\left[\sum_{t=1}^{T} A_{t,i}^2 \,\middle|\, V_{<i} = u\right],
\end{aligned}$$

where the final expectation is under the original hypercube prior and the algorithm's trajectory law. Averaging over $V_{<i}$ and summing over i gives

$$\begin{aligned}
I(V; H_T) &\le \Delta^2\mathbb{E}\sum_{t=1}^{T}\sum_{i=1}^{d} A_{t,i}^2 \\
&= \Delta^2\mathbb{E}\sum_{t=1}^{T}\|A_t\|_2^2 \\
&\le \Delta^2 T.
\end{aligned}$$

This proves the claim. □

Remark 7.7 (Why the fixed-design log-determinant argument is not used)**.** *For a Gaussian prior on a full parameter vector, posterior conjugacy gives the adaptive information identity*

$$I(\Theta; H_T) = \frac{1}{2}\mathbb{E}\log\det\left(I + \Sigma_0^{1/2}\Big(\sum_{t=1}^{T} A_t A_t^\top\Big)\Sigma_0^{1/2}\right),$$

and this is bounded by a trace-constrained log determinant. For the hypercube prior, however, the posterior is not Gaussian and the realized design matrix is itself a statistic of the observations. Conditioning only on that realized design matrix controls the conditional observation information given the design; it does not by itself control the information carried by the adaptive design. Lemma 7.6 therefore uses a coordinatewise adaptive KL argument to control $I(Y; H_T)$ directly.

Theorem 7.8 (Linear bandit lower bound). *For stochastic linear bandits with action set $\{a \in \mathbb{R}^d : \|a\|_2 \le 1\}$, unit Gaussian noise, and unit-ball parameter class $\Omega_{\mathrm{lin}} = \{\omega \in \mathbb{R}^d : \|\omega\|_2 \le 1\}$,*

$$\mathfrak{R}_T^\star \ge c\, d\sqrt{T}$$

for a universal constant $c > 0$, whenever $T \ge d^2$.

Proof. Assume first that d is larger than a sufficiently large universal constant d_0 to be fixed below. Choose

$$\Delta = c_1\sqrt{\frac{d}{T}},$$

where $c_1 > 0$ is a sufficiently small universal constant. Since $T \ge d^2$, choosing $c_1 \le 1$ ensures

$$\Delta\sqrt{d} = c_1\frac{d}{\sqrt{T}} \le c_1 \le 1,$$

so the hypercube prior is supported on the unit parameter ball.

Let

$$r = \frac{\Delta\sqrt{d}}{4}.$$

Lemma 7.5 gives the reference ghost-good probability bound

$$p_r \le \exp(-c_{\mathrm{gh}}d).$$

Lemma 7.6 gives the adaptive action-index information bound

$$T\bar{C} = I(Y; H_T) = I(V; H_T) \le \Delta^2 T = c_1^2 d.$$

Choose c_1 so small that $c_1^2 \le c_{\mathrm{gh}}/4$, and then choose d_0 so large that $\log 2 \le c_{\mathrm{gh}}d/4$ for all $d \ge d_0$. Then, for all $d \ge d_0$,

$$T\bar{C} + \log 2 \le \frac{1}{2}\log\frac{1}{p_r}.$$

Theorem 6.1, applied to the action index $Y = \chi(\Omega) = A^\star(\Omega)$, gives a Bayes regret lower bound under the hypercube prior of at least

$$\frac{Tr}{2} = \frac{T\Delta\sqrt{d}}{8} = \frac{c_1}{8}d\sqrt{T}.$$

Since this prior is supported on Ω_{lin}, the minimax regret over Ω_{lin} is at least this Bayes risk.

It remains to handle the finitely many dimensions $1 \le d < d_0$. For these dimensions, use the two-point prior $\Omega = \sigma\delta e_1$, where $\sigma \sim \mathrm{Unif}(\{\pm 1\})$, e_1 is the first coordinate vector, and $\delta = (2\sqrt{T})^{-1}$. This prior is also supported on the unit parameter ball. Under $\sigma = +1$, the optimal action is e_1; under $\sigma = -1$, the optimal action is $-e_1$. Let $\mathbb{P}_+$ and $\mathbb{P}_-$ be the laws of the history under these two environments, and let $\mathbb{P}_\pm^{t-1}$ denote the law of the history before action t. For any algorithm,

$$\begin{aligned}\frac{1}{2}\mathbb{E}_+\big[\delta(1 - A_{t,1})\big] + \frac{1}{2}\mathbb{E}_-\big[\delta(1 + A_{t,1})\big] &= \delta\left(1 - \frac{1}{2}(\mathbb{E}_+A_{t,1} - \mathbb{E}_-A_{t,1})\right)\\ &\ge \delta\left(1 - \mathrm{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1})\right).\end{aligned}$$

By Pinsker's inequality and the adaptive Gaussian KL chain rule,

$$\mathrm{TV}(\mathbb{P}_+^{t-1}, \mathbb{P}_-^{t-1}) \le \sqrt{\frac{1}{2}D_{\mathrm{KL}}(\mathbb{P}_+^{t-1}\|\mathbb{P}_-^{t-1})} \le \delta\sqrt{t-1} \le \frac{1}{2}.$$

Therefore each round has Bayes regret at least $\delta/2$, and the total Bayes regret is at least $T\delta/2 = \sqrt{T}/4$. Since $d < d_0$, this is at least $(4d_0)^{-1}d\sqrt{T}$. Reducing the universal constant c, if necessary, completes the proof for all d. $\square$

7.4 Extensions to contextual bandits and reinforcement learning

There are two rigorous but different ways to use contextual-bandit and finite-horizon RL estimators in this framework. The exact route is a belief-state lift: include the current context or physical state, the inner Bellman stage, and a coherent reference belief whose update is closed. The frequentist route is a calibrated certificate: include the estimator, design matrix, confidence set, relaxation, or domination object needed to upper-bound the fixed-truth Bellman bracket. IPW estimators, least-squares operators, and Bellman-rank witnesses are therefore useful stagewise coordinates, but they are not automatically complete Bellman-sufficient states. In episodic RL the Bellman time is the micro-time (k, h), where k is the outer learning episode and $h \in [H]$ is the inner horizon stage.

Example 7.9 (Adversarial contextual bandits and IPW estimator states)**.** *Consider an oblivious adversarial contextual bandit with finite action set $[K]$, policy class Π, contexts x_t, and losses $\ell_t(a) \in [0,1]$. The fixed truth is the whole sequence*

$$\omega^\star = (x_t, \ell_t(1), \ldots, \ell_t(K))_{t \le T}.$$

At the decision time for round t, the current context x_t is part of the state. The observation after acting may be written as $O_t = (\ell_t(A_t), x_{t+1})$, with the next context omitted at $t = T$. This convention makes the transition $S_{t+1} = \tau_t(S_t, A_t, O_t)$ closed. Given an action distribution $p_t(\cdot \mid x_t)$, the inverse-probability-weighted policy-loss increment

$$\widehat{\ell}_t(\pi) = \frac{\ell_t(A_t)\, \mathbb{1}\{\pi(x_t) = A_t\}}{p_t(A_t \mid x_t)}$$

satisfies $\mathbb{E}[\widehat{\ell}_t(\pi) \mid H_{t-1}, x_t, \omega^\star] = \ell_t(\pi(x_t))$ whenever the denominator is positive. Thus the cumulative vector $\widehat{L}_t(\pi) = \sum_{s<t} \widehat{\ell}_s(\pi)$ is a valid loss-estimation coordinate for relaxation-based contextual-bandit algorithms such as BISTRO (Rakhlin and Sridharan, 2016) and for mirror-descent information-ratio analyses (Lattimore and György, 2021). For an infinite policy class, $\widehat{L}_t$ should be understood as the exact functional $\pi \mapsto \widehat{L}_t(\pi)$, or as an oracle representation sufficient to evaluate the relaxation; a finite-dimensional sketch is sufficient only when the relaxation can be recovered from that sketch up to charged error.

It is not, however, an exact indexed Bellman state by itself. There are two rigorous augmentations. In the exact belief-state lift, one uses

$$S_t = (t, x_t, \widehat{L}_t, p_t, b_t),$$

where b_t is a reference law over the remaining context/loss sequence or over a parametrized environment class. Then the reference predictive law, the conditional index predictive laws, and the indexed marginal $q_t = \chi_\# b_t$ are determined by S_t, so Definition 2.2 applies to the reference experiment. In the frequentist certificate route, one uses

$$S_t = (t, x_t, \widehat{L}_t, p_t, C_t),$$

where C_t is the relaxation, confidence, or domination object proving that the IPW loss estimates upper-bound the fixed-truth Bellman bracket. This could verify the route of Definition 7.12 (see Section 7.6 for details), though not the straightforward posterior clauses of Definition 2.2. In both routes the IPW vector is an update coordinate; it is not the whole state. Unbiasedness identifies conditional expected losses, while Bellman sufficiency additionally requires closed prediction or certified domination, closed updating, and compatible index accounting.

The purpose of the following linear MDP example is to explain why the usual least-squares state for linear MDPs is not, by itself, naturally Bellman-sufficient, and how a finer specification guided by Bellman-sufficient states can make the horizon-wise and dimension-dependent structure across inner stages more explicit. This perspective complements standard episodic analyses, which are highly effective for regret bounds but often place less emphasis on specifying the sufficient state needed for an effective dynamic-programming recursion.

Example 7.10 (Linear MDPs and stagewise least-squares Bellman states)**.** *Consider an episodic finite-horizon MDP with outer episodes $k = 1, \dots, K$ and inner stages $h = 1, \dots, H$. The fixed truth $\omega^\star$ consists of the stagewise reward and transition kernels $(r_h^\star, P_h^\star)_{h=1}^H$. In the micro-time convention, the Bellman decision time is (k, h): the retained state contains the current physical state $X_{k,h}$ and the inner stage h, the action is $A_{k,h}$, and the observation is*

$$O_{k,h} = (R_{k,h}, X_{k,h+1}),$$

with the convention that after $h = H$ the next retained state begins episode $k + 1$ at stage 1. The outer episode counter k records the amount of data collected across learning episodes.

Suppose that there are known features $\phi_h(x, a) \in \mathbb{R}^d$ such that rewards and transition expectations are linear in ϕ_h. For a candidate next-value function V_{h+1}, the stage-h Bellman target has the form

$$r_h^\star(x, a) + \mathbb{E}_{P_h^\star(\cdot|x,a)}[V_{h+1}(X')] = \phi_h(x, a)^\top w_h^\star(V_{h+1}).$$

Least-squares value iteration is a stagewise estimation procedure, but the fitted estimator is not automatically a stagewise Bellman-sufficient state. For algorithms that update between episodes, the stage-h data available at the start of episode k gives the operator

$$\Lambda_{k,h} = \lambda I + \sum_{i<k} \phi_h(X_{i,h}, A_{i,h})\phi_h(X_{i,h}, A_{i,h})^\top,$$

$$u_{k,h} = \sum_{i<k} \phi_h(X_{i,h}, A_{i,h})R_{i,h}, \qquad \mathcal{M}_{k,h} : V \mapsto \sum_{i<k} \phi_h(X_{i,h}, A_{i,h})V(X_{i,h+1}),$$

so that

$$\widehat{w}_{k,h}(V) = \Lambda_{k,h}^{-1}\{u_{k,h} + \mathcal{M}_{k,h}V\}.$$

If the algorithm updates within an episode, the same display is modified by including the observations that are available before the decision time (k, h); the sufficiency requirement is unchanged.

The least-squares vector $\widehat{w}_{k,h}(V)$ is sufficient only for the single backup using that particular V. It is not, by itself, a Bellman-sufficient representation for the full planning recursion, because later or earlier dynamic-programming backups may use different continuation value functions. A Bellman-sufficient state for an exact reference experiment must include enough information to compute the relevant operator $V \mapsto \widehat{w}_{k,h}(V)$ for every continuation value used by the maintained value class or admissible relaxation, together with $(k, h, X_{k,h})$ and a coherent reference belief over the reward and transition parameters. A calibrated frequentist LSVI state may instead include the stagewise Gram matrices, target operators, computed value functions, and confidence radii that certify Bellman-target errors uniformly over the maintained class.

The examples have two consequences for later applications. First, exact Bayesian posterior states, Gaussian marginal states, finite conjugate statistics, and coherent algorithmic-belief lifts directly satisfy Definition 2.2 for their corresponding reference experiments. In particular, Section 2.5 shows that the full environment posterior is a fallback sufficient state for episodic model learning. Second, estimator-based frequentist states, such as IPW estimators and LSVI operators, are not automatically sufficient. They become Bellman-sufficient only when paired with the calibration, reference belief, domination inequality, or evaluation oracle that makes the fixed-truth Bellman bracket a function of the retained state. Thus, specifying the sufficient state is itself part of deriving a more intrinsic complexity measure for sequential decision making.

7.5 Connection to Bellman rank

Low Bellman rank was introduced by Jiang et al. (2017) as a sufficient structural condition under which contextual decision processes with suitable bilinear Bellman-error structure are PAC-learnable. In the present language, a Bellman-rank factorization can help build a low-dimensional Bellman-sufficient representation when it is supplemented by predictive, update, loss, and index sufficiency; the factorization alone controls Bellman-error evaluation and does not, by itself, determine the full observation process. The following proposition is structural; it is not a new sharp lower bound for every Bellman-rank class.

Proposition 7.11 (Bellman rank with predictive sufficiency gives an indexed representation)**.** *Suppose a contextual decision process admits an exact Bellman-error factorization of rank d in the sense of Jiang et al. (2017): for the stages and hypothesis–policy pairs in the planning class, the relevant Bellman error can be written as an inner product of two d-dimensional witness-feature vectors. Fix a prior μ and an index map χ. If the observation model is dominated by a common σ-finite measure and if, conditional on the current posterior or calibrated confidence object, the predictive law, the posterior-reference update, the relevant loss or benchmark terms, and the conditional χ-indexed predictive laws and losses in Definition 2.2 are all determined by these Bellman-error features together with the retained planning variables, then the process admits an indexed Bellman sufficient representation with state that may be chosen as a posterior or calibrated confidence object over the d-dimensional witness coordinates, augmented by the planning variables needed for Bellman updates. Consequently, whenever one can evaluate or validly upper-bound the exact information-capacity and ghost-good-mass Bellman values for the induced feature experiment, for example in a linear-Gaussian or uniformly bounded-likelihood setting, Theorem 6.3 yields a Bellman quantile-index lower certificate.*

Proof. The Bellman-error factorization supplies d-dimensional coordinates sufficient to evaluate the one-step Bellman-error terms covered by the factorization. It does not by itself imply predictive sufficiency, loss sufficiency, update sufficiency, or index sufficiency. These are precisely the additional assumptions imposed in the proposition. If the predictive observation law and posterior update are functions of this statistic and the current posterior, and if the relevant losses, benchmarks, and χ-conditional predictive objects are functions of the same retained state, then Definition 2.2 is satisfied with S_t equal to the posterior or calibrated confidence object over these features together with the accumulated planning state. The information and ghost-good Bellman recursions are then well-defined on the same state. Applying Theorem 6.3 to their exact values, or to any valid pair of supersolutions for this state space, gives the claimed certificate. The proposition is intentionally conditional because Bellman-rank learnability is primarily a structural upper-bound statement; sharp lower bounds require a packing or ghost-entropy calculation inside the particular rank-d class. □

This application clarifies the connection between the present framework and the broader perspective on representation learning developed in Li and Xu (2026); Xu (2026). The point is not that every low-Bellman-rank problem has the same minimax rate or that Bellman rank alone determines a lower bound. The point is that a Bellman-rank factorization can provide a candidate state on which the upper log-potential certificate and the lower Bellman-Fano values should be compared, after the necessary predictive, update, loss, and index-sufficiency conditions have been verified.

7.6 Extension to frequentist estimators as Bellman-sufficient state

The exact Bellman-sufficiency definition in Definition 2.2 corresponds to the posterior-process/reference-law formulation. This formulation is designed to state the fundamental specification conditions and to support lower-bound analysis. In this formulation, posteriors induced by fixed algorithmic beliefs may be used as frequentist estimators and may qualify as Bellman-sufficient states when they close the required Bellman recursions. By contrast, estimator-based frequentist algorithms, such as IPW contextual bandits and least-squares value iteration, typically verify a calibrated certificate rather than exact posterior-predictive sufficiency. The same is true for the adaptively optimized algorithmic posteriors of Xu and Zeevi (2025); Liu et al. (2025, 2026), where the posterior process is chosen for algorithmic or certification purposes rather than inherited from a fixed Bayesian belief. The following more flexible certificate is based on the upper-bound theory and notation of Section 5, and extends the spirit of Definition 2.2 to common frequentist estimators beyond algorithmic posteriors.

Definition 7.12 (Calibrated fixed-truth Bellman certificate). *Fix a class $\Omega_0 \subseteq \Omega$, an index χ, an indexed marginal q_s whenever a logarithmic information potential is used, and a potential sequence Ψ_t. A state process S_t is a calibrated fixed-truth Bellman certificate on Ω_0 if it satisfies the closed action/update requirements and, for each state s, determines a comparison set $\mathfrak{C}_t(s) \subseteq \Omega_0$ or a reference belief b_s, together with state-measurable quantities sufficient to evaluate or upper-bound the robust fixed-truth bracket*

$$\mathcal{B}_t^{\Psi,\gamma}(s,p) := \sup_{\omega \in \mathfrak{C}_t(s)} \Big\{ \ell_\omega(s,p) + \mathbb{E}_{a\sim p,\, O\sim P_{\omega,s,a}} \Psi_{t+1}(\tau(s,a,O)) - \Psi_t(s) - \gamma\, \mathsf{G}_\chi(s,p;\omega) \Big\},$$

where

$$\mathsf{G}_\chi(s,p;\omega) = \mathbb{E}_{a\sim p,\, O\sim P_{\omega,s,a}} \log \frac{q_{\tau(s,a,O)}(\chi(\omega))}{q_s(\chi(\omega))}$$

when an indexed belief q_s is maintained; if no such belief is used, the last term is replaced by the state-measurable uncertainty or relaxation increment appearing in the chosen supersolution. Calibration means that, for every fixed truth $\omega^\star \in \Omega_0$, either $\omega^\star \in \mathfrak{C}_t(S_t)$ on the relevant good event or the failure probability and approximation error are explicitly charged, and the actual conditional Bellman bracket under $\omega^\star$ is bounded by $\mathcal{B}_t^{\Psi,\gamma}(S_t,p_t)$ plus those charged errors.

Thus a sufficient state for calibrated Bellman certificate need not be an exact posterior state, but it must contain enough information to make the fixed-truth Bellman inequality a function of the state. Unbiasedness, least-squares normal equations, or efficiency of an estimator are useful only insofar as they establish this calibration condition.

8 Conclusion

This paper is organized around a single principle: the statistical difficulty of an interactive decision problem is determined only after one has chosen the right sufficient state and the right index. The environment remains the fixed frequentist truth that generates observations and losses. The Bellman-sufficient state is the part of the history on which prediction, loss evaluation, updating, and future control can be continued. The index $Y = \chi(\Omega)$ is the object whose identification is charged. It may be the optimal action, the optimal policy, a value object, an active finite marginal, or the full model; choosing it is part of the representation. This separation is the main conceptual point. It prevents an upper bound from paying for irrelevant model estimation, and it prevents a lower bound from proving hardness for information that the decision problem never requires.

On the upper side, the basic object is not a one-step coefficient, but a Bellman supersolution with a logarithmic information coordinate. For a fixed truth $\omega^\star$ and $y^\star = \chi(\omega^\star)$, the coordinate potential

$$\gamma \log \frac{1}{q_t(y^\star)}$$

turns indexed learning into a dynamic program: regret is controlled by immediate loss, continuation value, and the fixed-truth log gain of the maintained reference marginal. The exact log-penalized Bellman program is therefore the canonical information-theoretic upper algorithm. Its posterior-averaged form recovers Bayesian AIR/MAIR identities, but the fixed-truth form is more fundamental for minimax analysis. UCB, E2D, and AMS/EBO are best understood as tractable ways to certify or relax this same Bellman bracket. UCB uses calibration and optimism to upper-bound the fixed-truth bracket; E2D keeps the immediate loss and a one-round separation penalty while dropping or freezing the continuation value; AMS/EBO replaces the exact continuation value by a KL-dual log-partition, or admissible-relaxation, potential over calibrated candidate beliefs.

On the lower side, the same representation supports a Bellman-Fano value comparison. The information term is the exact or certified indexed-information Bellman capacity along posterior-reference trajectories, while the entropy term is the exact or certified ghost-good mass of reference histories whose target-wise regret is small. Thus the lower bound does not ask how many environments can be distinguished in the full model class. It asks how much indexed information the interactive experiment can reveal before it would make too many low-regret ghost histories possible. Theorem 3.4 expresses the resulting matching principle: when the log-penalized Bellman upper value and the Bellman-Fano ghost-quantile lower certificate close at the same radius on the same state/index representation, the minimax rate is determined at that radius. In this sense, the paper gives an interactive analogue of information-risk matching in fixed statistical experiments: local prior mass is replaced by ghost-good mass, and fixed-design KL is replaced by a controlled Bellman information telescope.

Several problems remain open. One is to identify, for broad classes of interactive problems, minimal or near-minimal Bellman-sufficient states. Another is to understand when convex belief relaxations such as AMS/EBO are tight approximations to the exact log-penalized Bellman program, rather than merely safe relaxations. A third is to develop lower certificates whose ghost-good entropy is as constructive as the corresponding upper algorithm. A fourth is to extend the same state/index discipline to misspecified, computationally constrained, and long-horizon reinforcement-learning settings where exact sufficiency is unavailable and every theorem must carry approximation error explicitly.

The conclusion is therefore methodological as much as technical. To analyze an interactive learning problem, one should not begin by asking for the dimension of the model, or for a universal conversion mechanism between estimation and decision. One should ask: what state representation makes the Bellman recursion close, what index is actually worth learning, what logarithmic potential pays for that index, and what ghost quantile witnesses the matching lower entropy? When these four objects align, learning becomes a special dynamic program whose value coordinate is information itself. The statistical complexity of the problem is then no longer a property of the raw model class alone; it is a property of the Bellman-sufficient representation through which the learner, the upper certificate, and the lower certificate all see the problem.

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